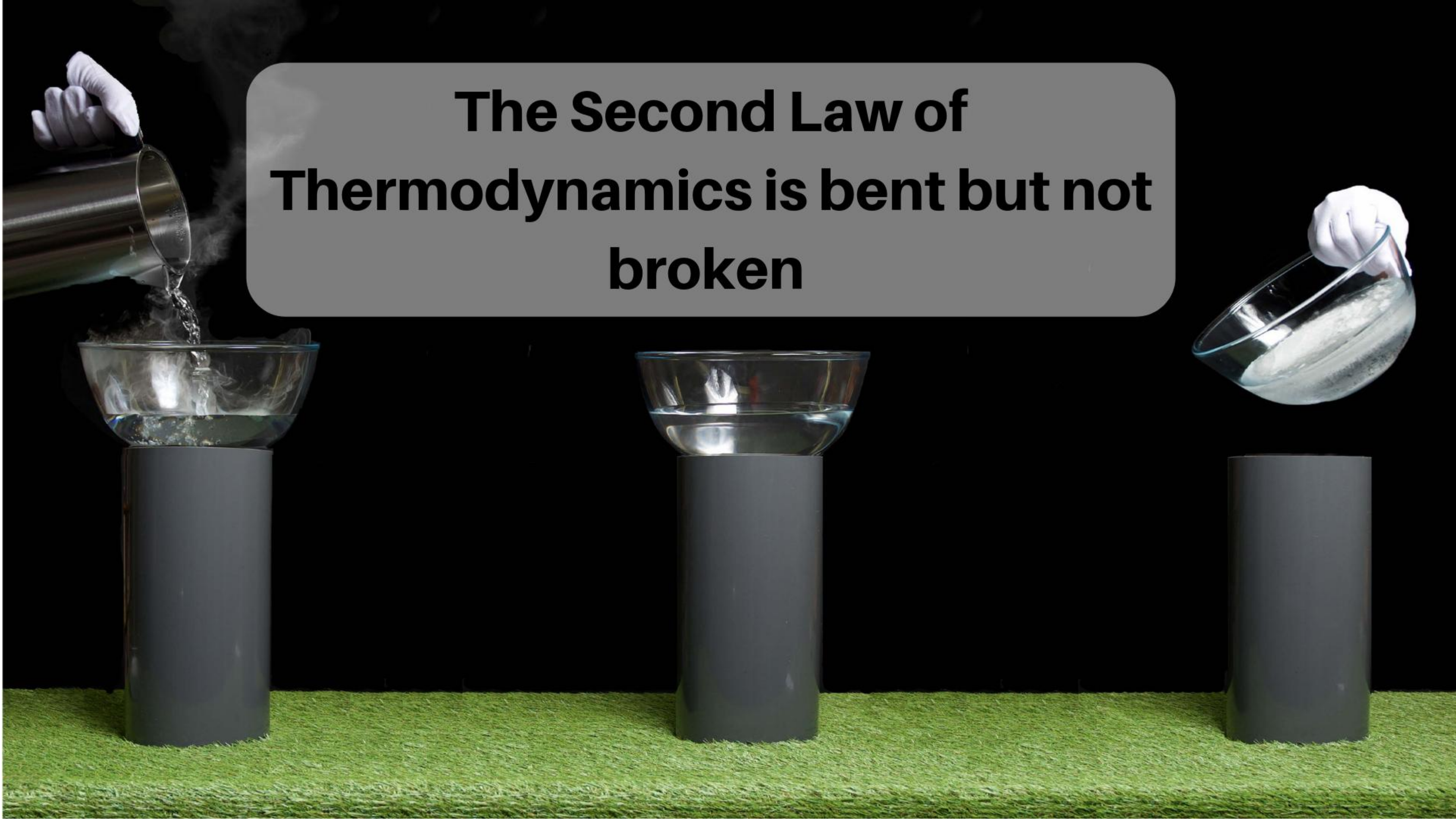


Module-4: Second Law of Thermodynamics

Thermodynamics (NMEC103) : (Winter 2024-25)

Dr. S. Sahoo, Mechanical Engineering, IIT Dhanbad

Some relations that have been developed for reversible processes involve only properties of the system and consequently are valid for processes that are internally reversible even if they are externally irreversible. As an example a gas is expanding in a quasi equilibrium process in a cylinder against a moving piston may have heat added across a finite temperature difference

$$w_{rev,nonflow} = \int Pdv$$

Analogous relation for a steady flow system

$$w_{rev,flow} = -\int vdP - \Delta\left(\frac{V^2}{2}\right) - g\Delta z$$

The external irreversibility in no way affects the pv relationship so the above relations still hold

$$w_{\text{int}\,rev,nonflow} = \int Pdv \qquad w_{rev,flow} = -\int vdP - \Delta\left(\frac{V^2}{2}\right) - g\Delta z$$

Similarly in entropy chapter it will be shown that for a reversible process either for a closed system or steady flow open system

$$Q_{rev} = \int T ds$$

Based on the similar reasoning it can be concluded that the above equation is valid even for process which is at least internally reversible i.e.

$$Q_{\mathrm{int},rev} = \int T ds$$

Cause of irreversibility:

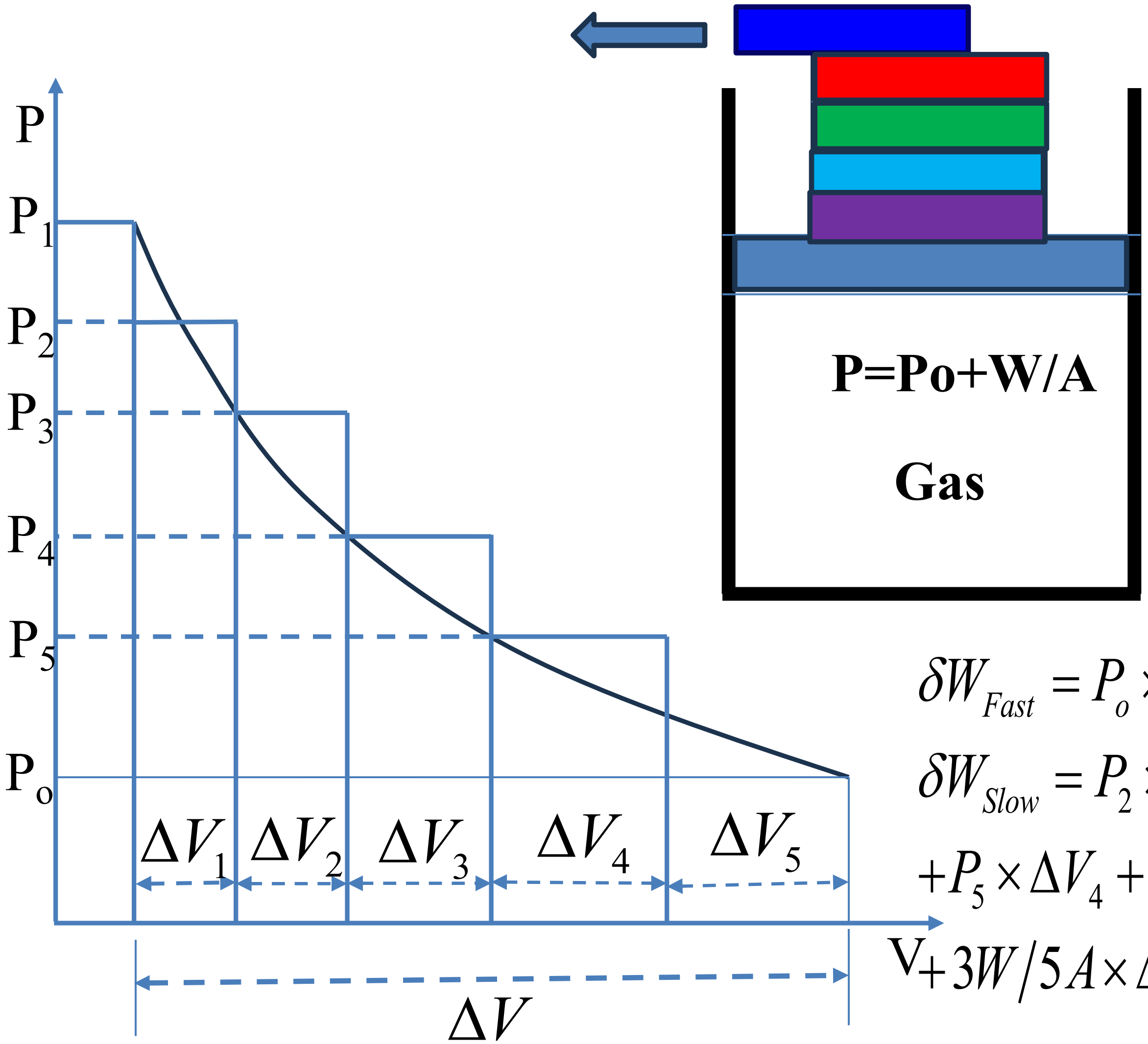
Lack of equilibrium (mechanical, thermal, chemical equilibrium) between the systems, two parts of the system or between system and surrounding

- ❖ Heat transfer through finite temperature difference
- ❖ Change of state through finite pressure gradient
- ❖ Free expansion

Presence of dissipative effect: such as friction, viscosity, inelasticity, electric resistance, and magnetic hysteresis

Irreversibility means loss of work potential

How does fast process give rise to irreversibility, i.e., loss of work potential?



$$\Delta V = \Delta V_1 + \Delta V_2 + \Delta V_3 + \Delta V_4 + \Delta V_5$$

$$P_2 = P_o + 4W/5A$$

$$P_3 = P_o + 3W/5A$$

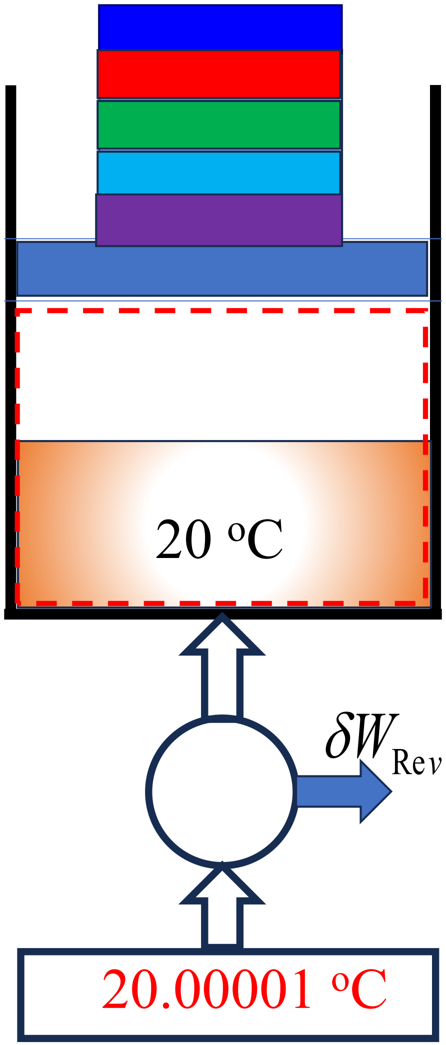
$$P_4 = P_o + 2W/5A$$

$$P_5 = P_o + W/5A$$

$$\delta W_{Fast} = P_o \times \Delta V$$

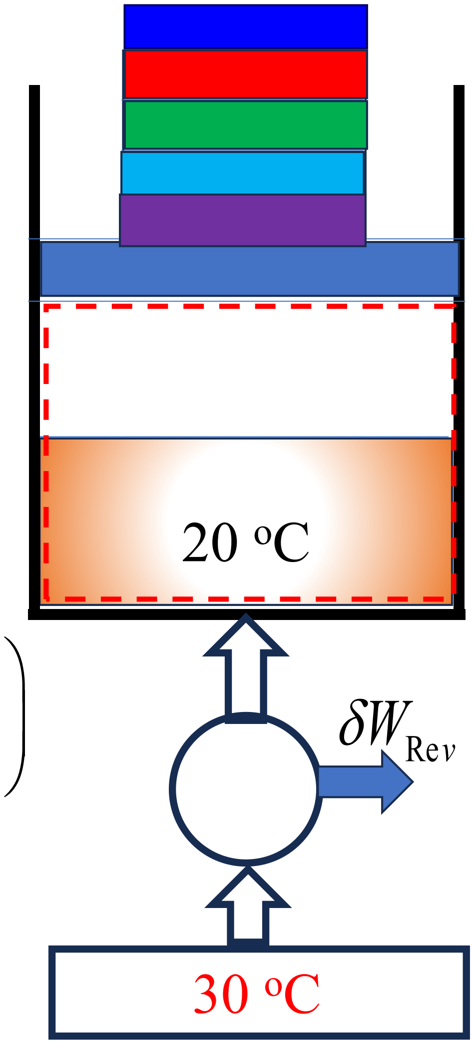
$$\delta W_{Slow} = P_2 \times \Delta V_1 + P_3 \times \Delta V_2 + P_4 \times \Delta V_3 + P_5 \times \Delta V_4 + P_o \times \Delta V_5 = P_o \times \Delta V + 4W/5A \times \Delta V_1 + 3W/5A \times \Delta V_2 + 2W/5A \times \Delta V_3 + W/5A \times \Delta V_4$$

How does heat transfer through large ΔT give rise to large irreversibility?



$$\delta W_{Rev} = \delta Q\left(1 - \frac{20+273}{20.00001+273}\right)$$

Lost Work is Less



$$\delta W_{Rev} = \delta Q\left(1 - \frac{20+273}{30+273}\right)$$

Lost Work is High

In summary, irreversible processes normally include one or more of the following

irreversibilities:

I. Heat transfer through a finite temperature difference

II. Unrestrained expansion of a gas or liquid to a lower pressure

III. Spontaneous chemical reaction

IV. Spontaneous mixing of matter at different compositions or states

V. Friction—sliding friction as well as friction in the flow of fluids

VI. Electric current flow through a resistance

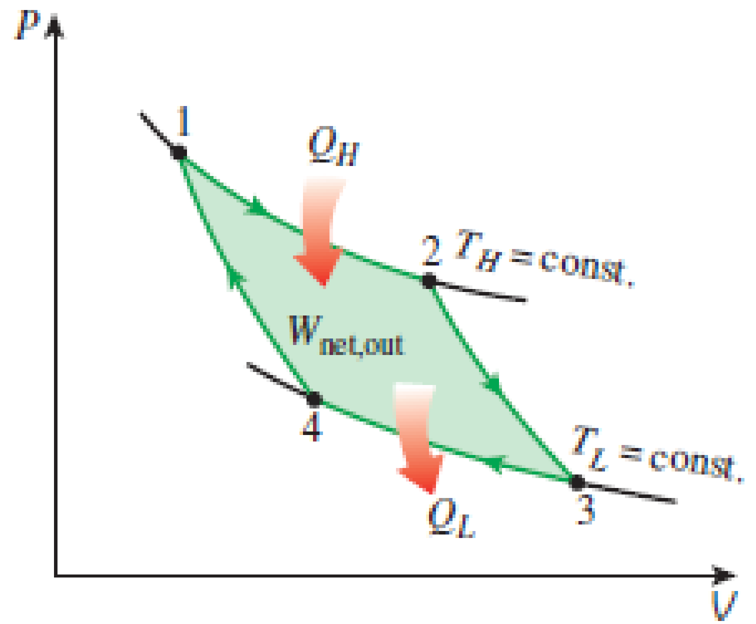
VII. Magnetization or polarization with hysteresis

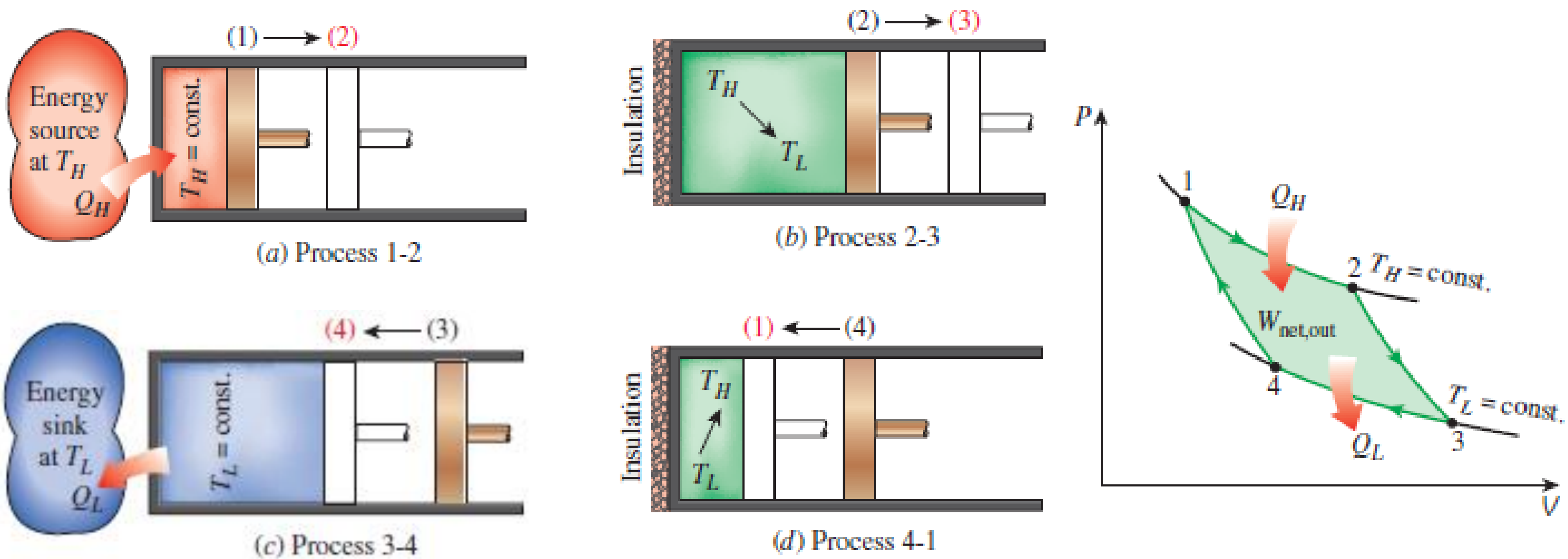
VIII. Inelastic deformation

Mixing is the best example of an irreversible process

Carnot Cycle

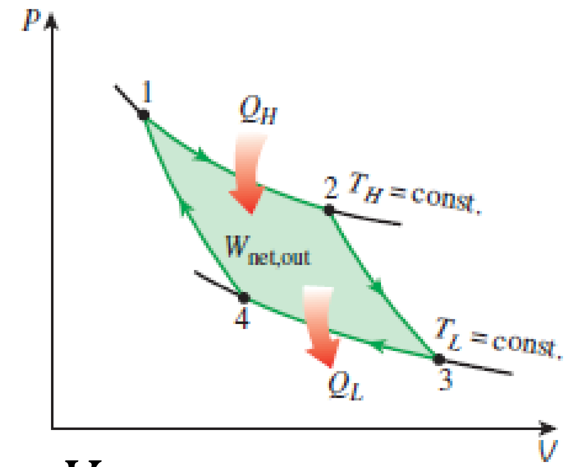
- The Carnot cycle is a fully reversible (both internally and externally) cycle comprising of two reversible adiabatic (2-3 & 4-1) and two reversible isothermal processes (1-2 & 3-4) (As discussed earlier the isothermal and reversible adiabatic processes are reversible both internally and externally).
- The isothermal heat addition and rejection made Carnot a fully reversible cycle as the source and sink are also assumed to be isothermal closed systems.
- If an engine operates on the Carnot cycle then the heat lost to the sink i.e. Q_L is not due to any irreversibility (dissipative effect) but because of the constraint put by second law i.e. heat can not be completely and continuously (i.e. in a cycle) converted into work.
- However, an irreversible heat engine operating between the same temperature limits and same heat input will reject more heat to the sink and produces less work. *This increase in heat loss or decrease in the work produced is because of the irreversibility.*





The Carnot cycle is not the only reversible heat-engine cycle. Two others are known as the Stirling and Ericsson Cycle.

For a Carnot engine using an ideal gas with constant specific heats as working fluid, derive an expression for thermal efficiency in terms of the reservoir temperature T_H and T_L



$$\eta = \frac{\oint \delta W}{Q_{in}} = \frac{\oint \delta Q}{Q_{in}} = \frac{Q_H - Q_L}{Q_H}$$

$$|Q_H| = U_2 - U_1 + W_{1-2}$$

$$|Q_H| = 0 + \int_1^2 PdV = mRT_H \int_1^2 \frac{dV}{V} = mRT_H ln\frac{V_2}{V_1}$$

$$|Q_L| = U_3 - U_4 - W_{3-4}$$

$$|Q_L| = 0 - \int_3^4 PdV = -mRT_L \int_3^4 \frac{dV}{V} = -mRT_H ln\frac{V_4}{V_3} = mRT_L ln\frac{V_3}{V_4}$$

$$\eta = 1 - \frac{Q_L}{Q_H} = 1 - \frac{|Q_L|}{|Q_H|} = 1 - \frac{mRT_L ln\frac{V_3}{V_4}}{mRT_H ln\frac{V_2}{V_1}}$$

For reversible adiabatic process of an ideal gas with constant specific heat

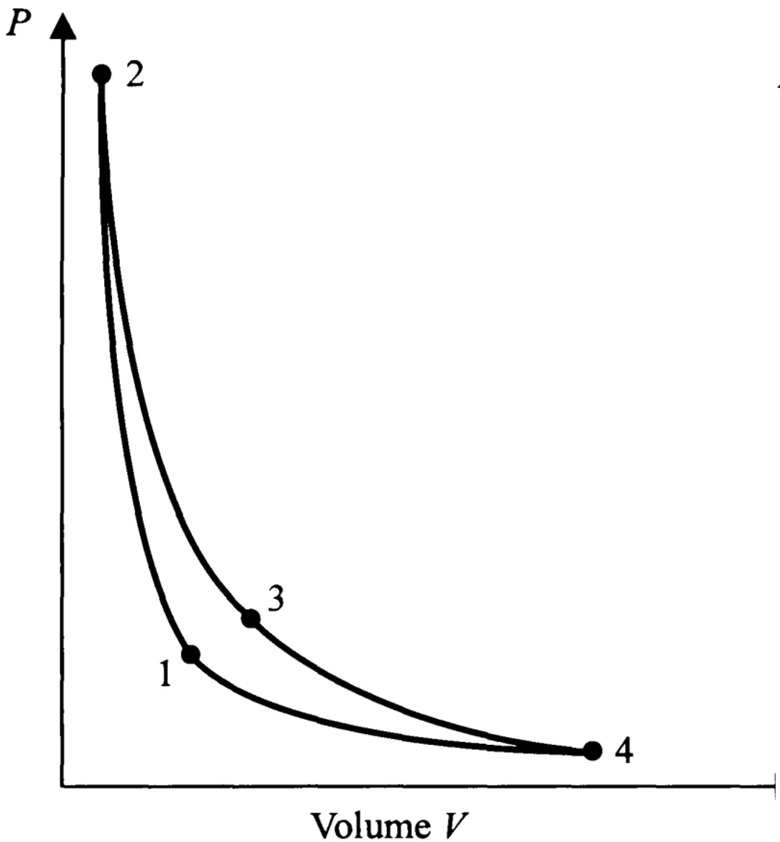
$$\frac{V_3}{V_2} = \left(\frac{T_2}{T_3}\right)^{1/(\gamma-1)} = \left(\frac{T_H}{T_L}\right)^{1/(\gamma-1)}$$

$$\frac{V_4}{V_1} = \left(\frac{T_1}{T_4}\right)^{1/(\gamma-1)} = \left(\frac{T_H}{T_L}\right)^{1/(\gamma-1)}$$

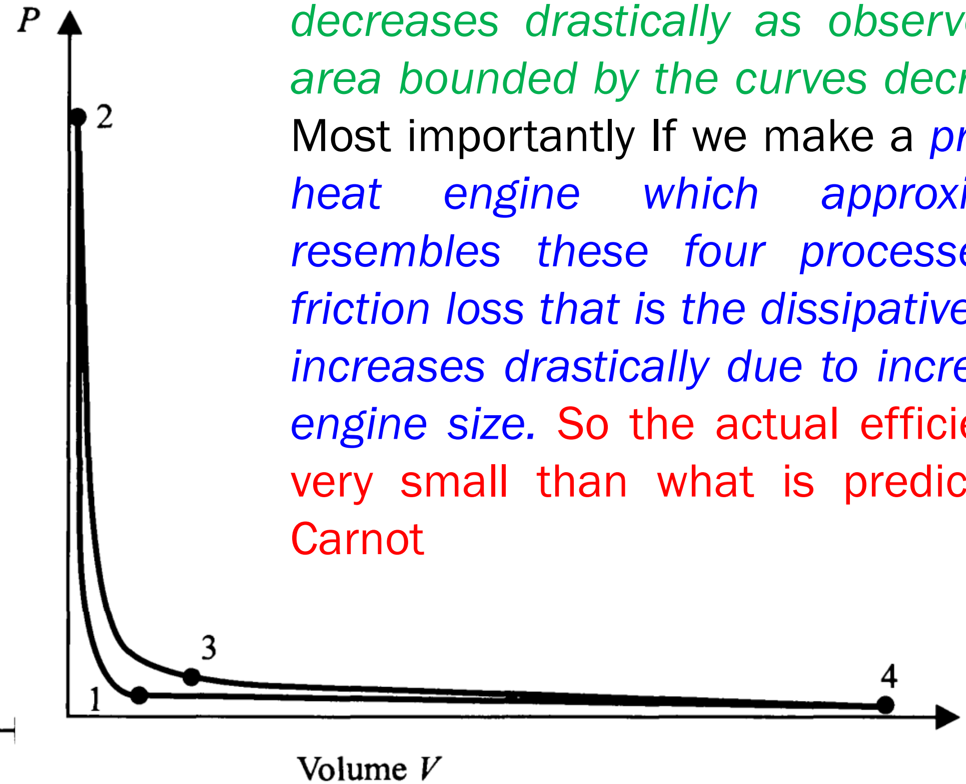
$$\frac{V_3}{V_2} = \frac{V_4}{V_1} \Longrightarrow \frac{V_3}{V_4} = \frac{V_2}{V_1}$$

$$\eta = 1 - \frac{Q_L}{Q_H} = 1 - \frac{|Q_L|}{|Q_H|} = 1 - \frac{mRT_L ln\frac{V_3}{V_4}}{mRT_H ln\frac{V_2}{V_1}} = 1 - \frac{T_L}{T_H}$$

Accurately Drawn Carnot Cycle on P-v Plot



$$\frac{V_3}{V_1} = \frac{3}{2}$$



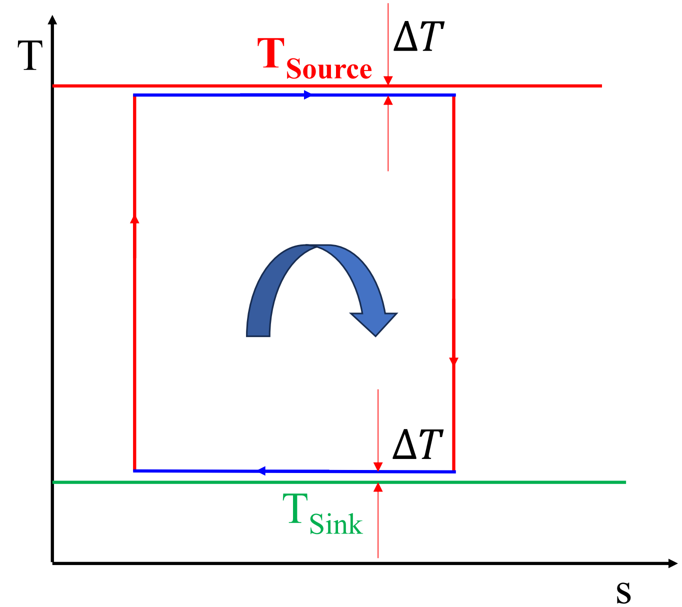
$$\frac{V_3}{V_1} = 3$$

If V_3/V_2 increases the Engine size increases and the *net work output decreases drastically as observed the area bounded by the curves decreases*. Most importantly If we make a *practical heat engine which approximately resembles these four processes the friction loss that is the dissipative effect increases drastically due to increase in engine size.* So the actual efficiency is very small than what is predicted by Carnot

Explanation, why Carnot is a fully reversible Cycle while Brayton, Rankine, Otto, Diesel etc. are only internally reversible

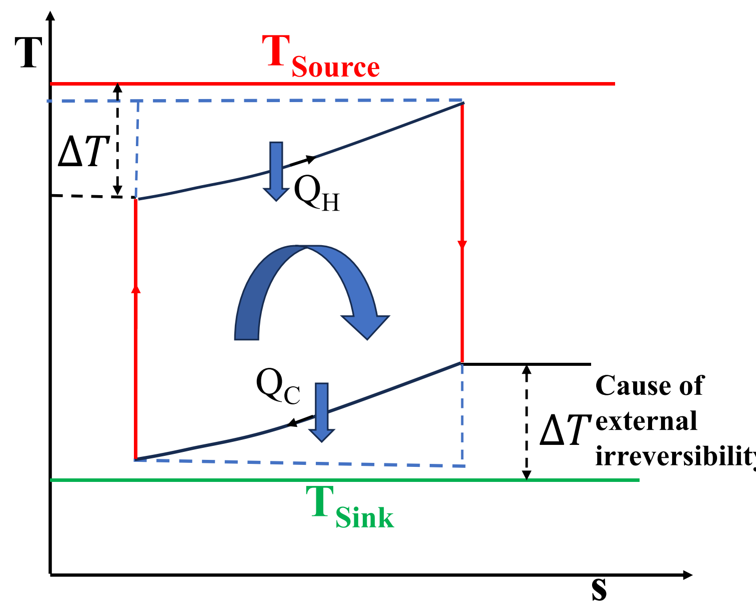
Both internally and externally reversible

Figure-1.: When source and sinks are constant temperature reservoirs, then isothermal heat addition/rejection allows the heat transfer process to be reversible by maintaining constant but infinitesimally small temperature difference between heat reservoirs and working fluid during heat addition/rejection process as shown in figure-1.



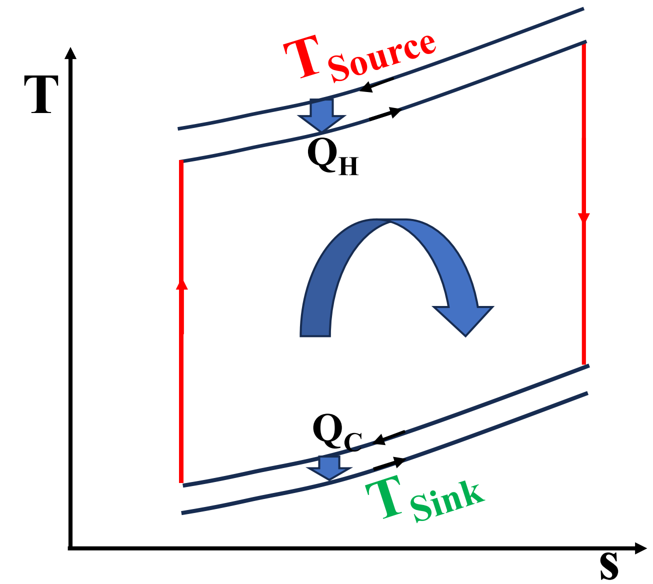
Only internally reversible

Figure-2 . In this case considering a counter flow heat exchanger it can be observed that although at one end we can have very low ΔT but ΔT increases to wards the other end of the heat exchanger along the length.



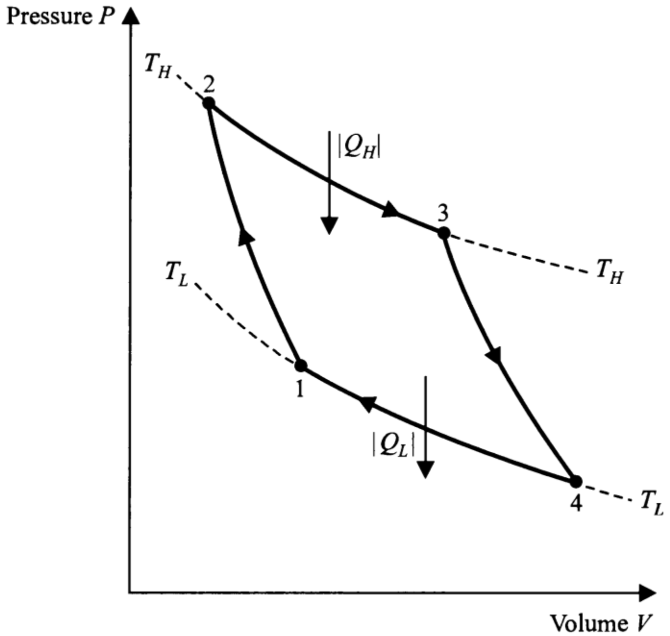
Both internally and externally reversible

Figure 3: Hence in those cases a cycle having heat addition process such that the temperature profile of the working fluid follow the same trend as the source and sink but in the opposite direction will result in a fully reversible cycle

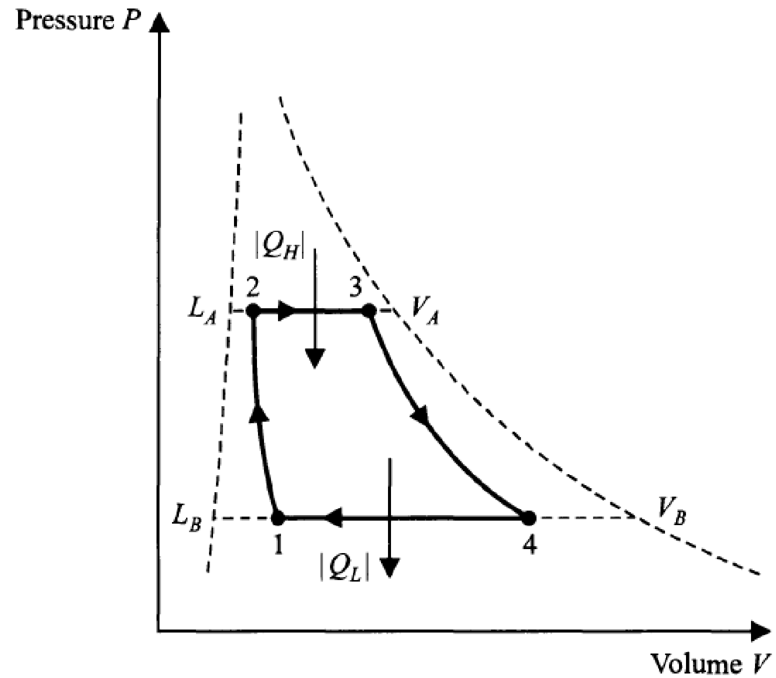


Example: In refrigeration when water is used as the cooling medium and refrigerated space temperature is also variable then a zeotropic mixture is used as the refrigerant, so that although constant pressure is maintained in the evaporator and condenser a temperature glide is observed during the phase change process. The resulting cycle is called the Lorenz cycle which results in better performance than the Carnot cycle

Examples of Carnot Cycle:



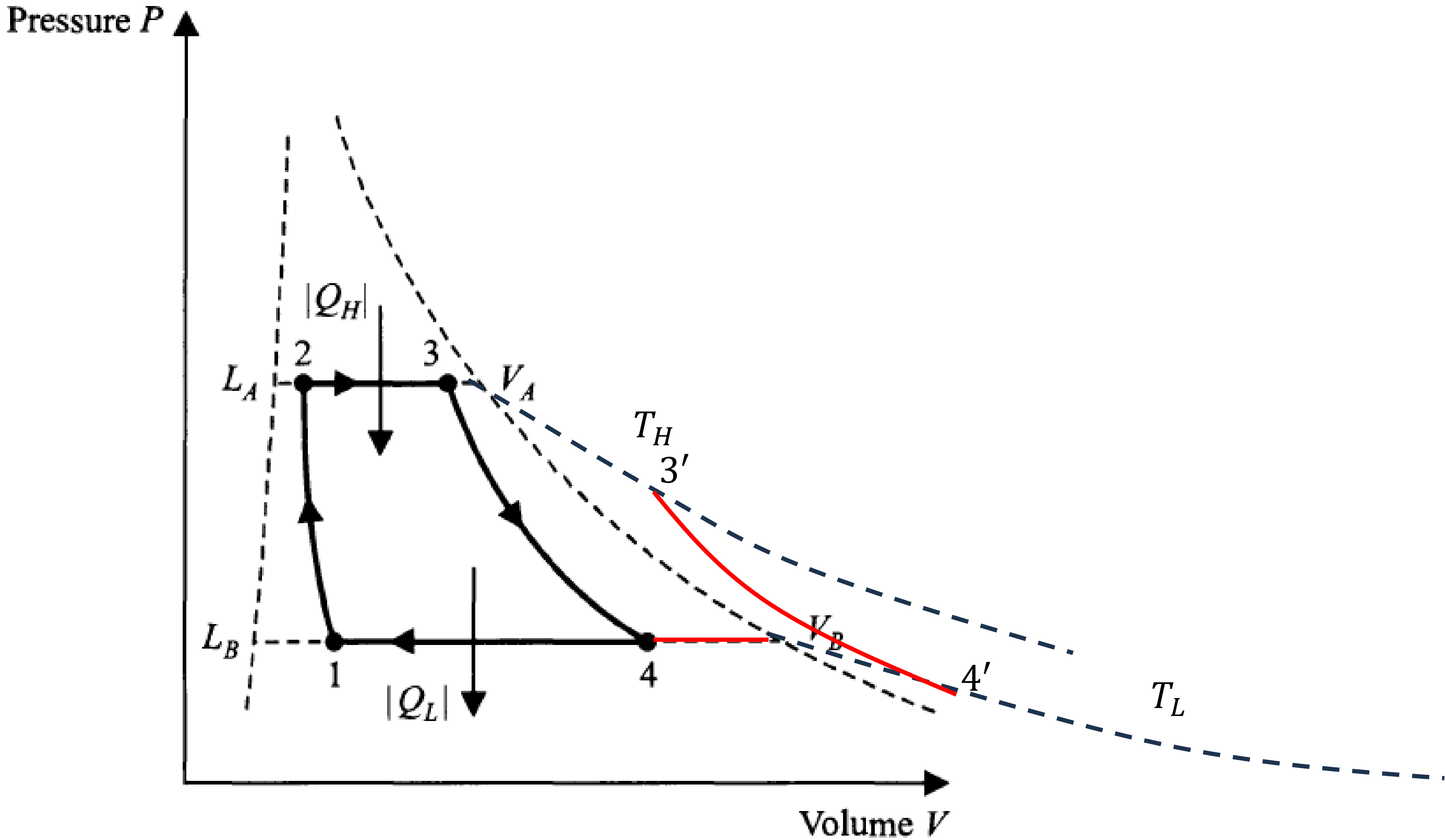
Carnot Cycle of Real gas



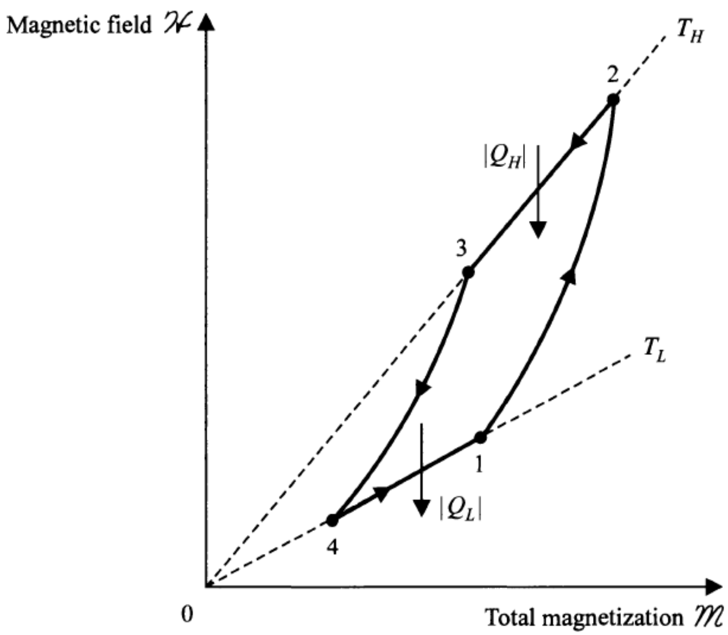
Carnot Cycle of mixture of liquid &vapour

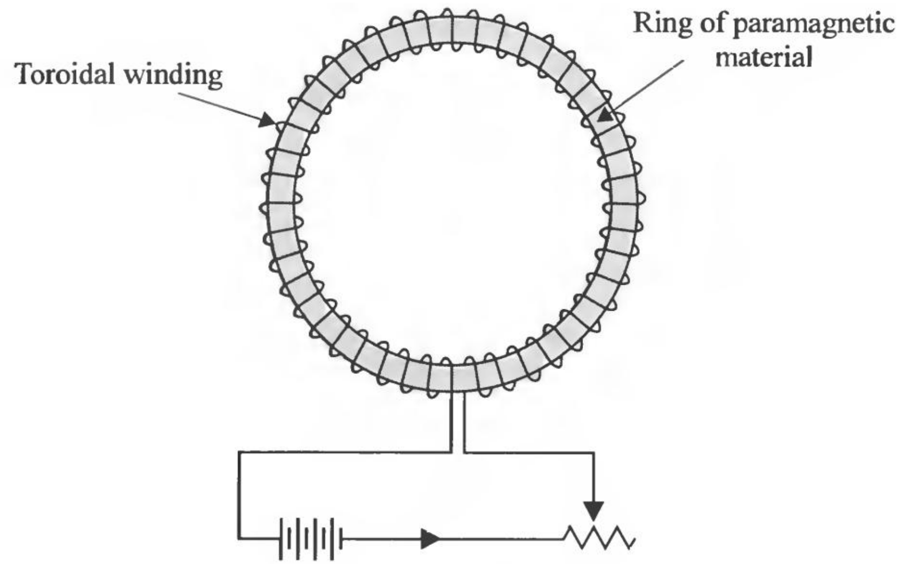
- 1-2 reversible adiabatic compression till the temperature rises to T_H
- Process 2-3 reversible isothermal expansion until point 3 is reached
- Process 3-4, reversible adiabatic expansion until temperature drops to T_L
- Process 4-1 reversible isothermal compression until initial state is restored

- 1-2 reversible adiabatic compression till the temperature rises to T_H
- Process 2-3 reversible isothermal & isobaric vaporisation until point 3 is reached
- Process 3-4, reversible adiabatic expansion until temperature drops to T_L
- Process 4-1 reversible isothermal isobaric condensation until initial state is restored


Pressure P
Volume V
$|Q_H|$
$|Q_L|$
L_A
L_B
V_A
V_B
T_H
T_L
1
2
3
4
3′
4′

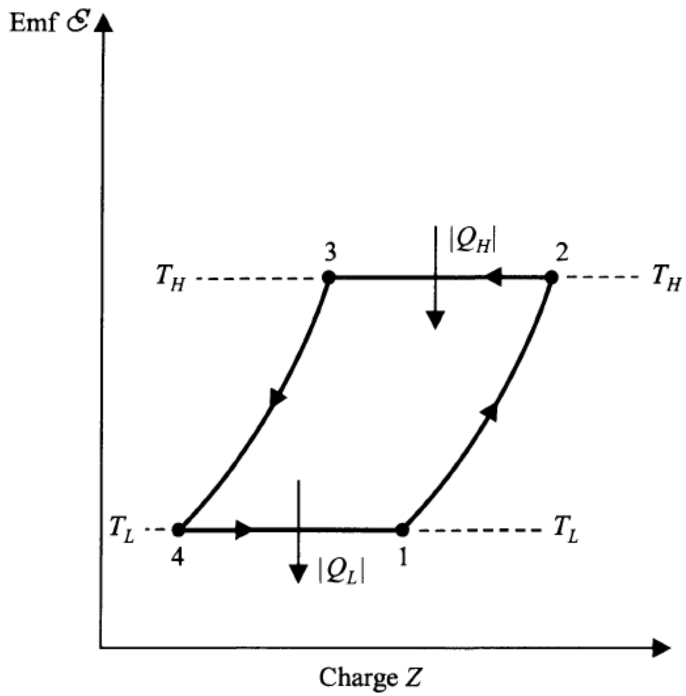
Carnot Cycle of a paramagnetic substance

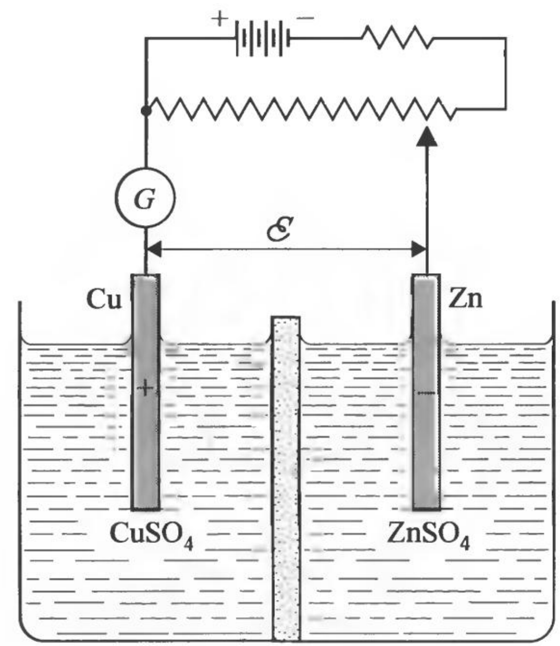




- Process 1-2 eversible adiabatic magnetisation until the temperature rises to T_H,
- Process 2-3 reversible isothermal demagnetisation until an arbitrary point 3 is reached,
- Process 3-4, reversible adiabatic demagnetisation until temperature drops to T_L,
- Process 4-1 reversible isothermal magnetisation until initial state is restored

Carnot Cycle of a Electro Chemical Cell





- 1-2 reversible adiabatic flow of charge from – to +ve in the external circuit till the temperature rises to T_H
- Process 2-3 reversible isothermal flow of charge from +ve to –ve in the external circuit until an arbitrary point 3 is reached
- Process 3-4, reversible adiabatic flow of charge until temperature drops to T_L
- Process 4-1 reversible isothermal flow of charge until initial state is restored

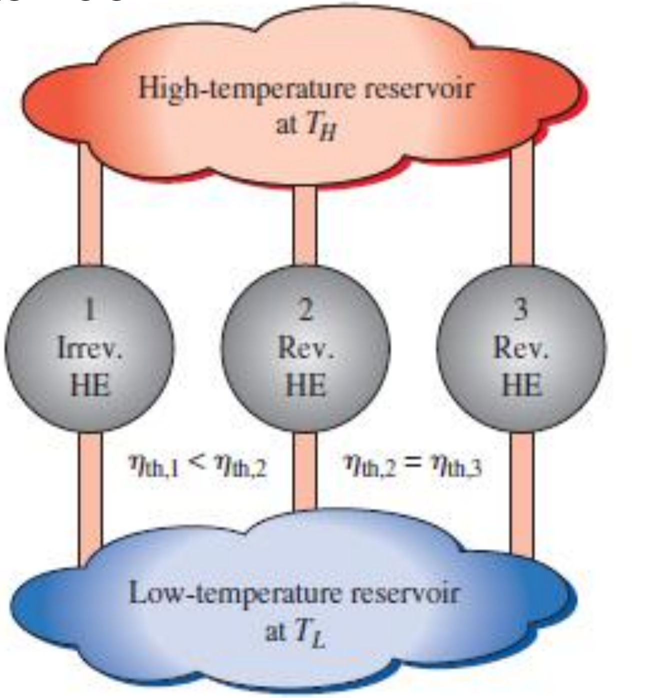
Carnot's Theorems

Theorem1: It is impossible to construct a heat engine that operates between two thermal reservoirs and is more efficient than a totally reversible engine operating between the same two reservoirs.

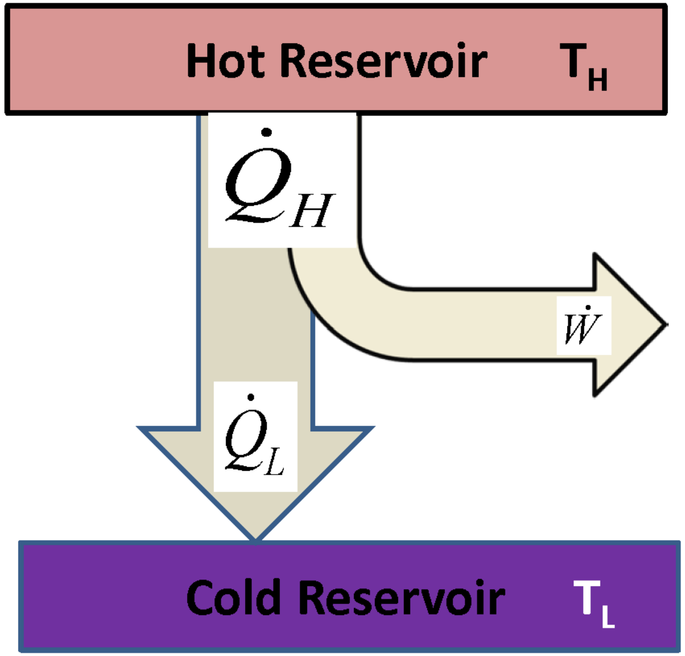
Theorem 2: All totally reversible heat engines operating between the same two thermal reservoirs have the same thermal efficiency.

Theorem-3: An absolute temperature scale may be defined which is independent of nature of the measuring substance.

Theorm-4 The efficiency of any reversible engine operating between more than two reservoirs must be less than that of a reversible heat engine operating between two reservoir which has temperature equal to the highest and lowest temperature of the fluid in the original engine.



Carnot's theorems

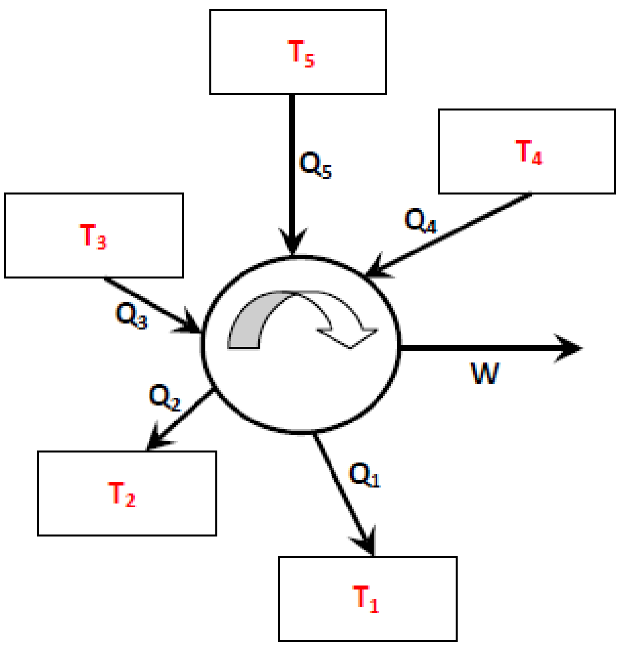


$$\eta_C = 1 - \frac{Q_L}{Q_H} = 1 - \frac{T_L}{T_H}$$

$$COP_{C,R} = \frac{T_L}{T_H - T_L} \qquad \frac{Q_L}{Q_H} = \phi(T_L, T_H)$$

$$COP_{C,H} = \frac{T_H}{T_H - T_L}$$

$$\frac{Q_L}{Q_H} = \frac{T_L}{T_H}$$ (For reversible case) $$\oint \frac{\delta Q}{T} = 0$$



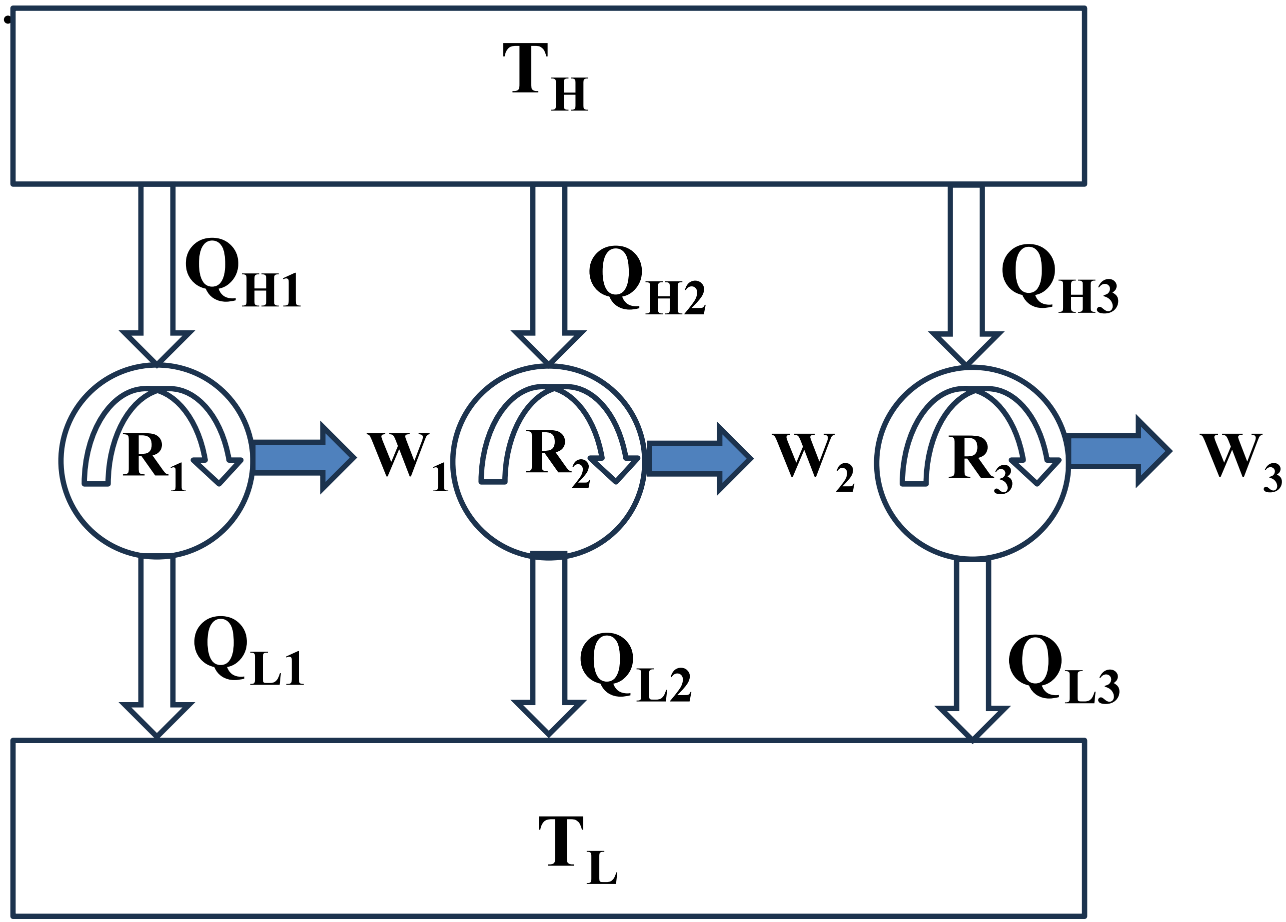
$$if\ T_1 < T_2 < T_3 < T_4 < T_5$$

$$\eta_C < 1 - \frac{T_1}{T_5}$$

Carnot's Theorems i.e., Law of equality and law of inequality

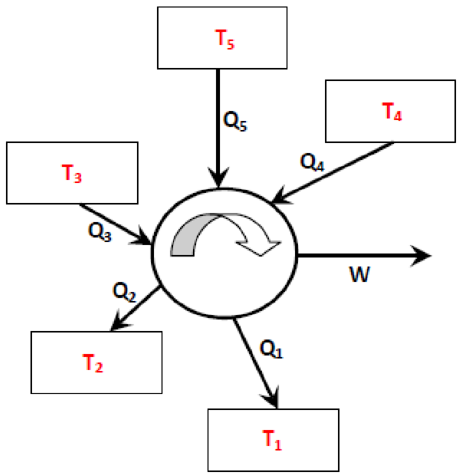
Law of Equality: Second Law also depicts that all fully reversible engines operating between same temperature limits have the same efficiency irrespective of the working substance and operating cycle.

$$\eta_{th,rev} = f(T_H, T_L)$$



$$\eta_{R1} = \eta_{R2} = \eta_{R3}$$

Law of Inequality: Second law talks about many inequalities (e.g. The performance of an irreversible engines/devices are less than that of the reversible ones operating between the same temperature limits). Second law introduces the famous inequality i.e. the Clausius inequality applicable for thermodynamic cycles



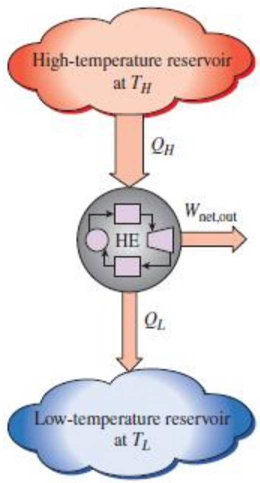
$$2nd\ Law: \sum_{i=1}^{i=n} \frac{Q_i}{T_i} \leq 0$$

Clausius inequality $\oint \frac{\delta Q}{T} \leq 0$

$$Entropy\ generation, S_{gen} = -\sum_{i=1}^{i=n} \frac{Q_i}{T_i} \geq 0$$

Law of Limits: Second law also put limits on the maximum efficiency or COP attainable by any heat engine and thermodynamic devices (Refrigerator &Heat Pump) respectively

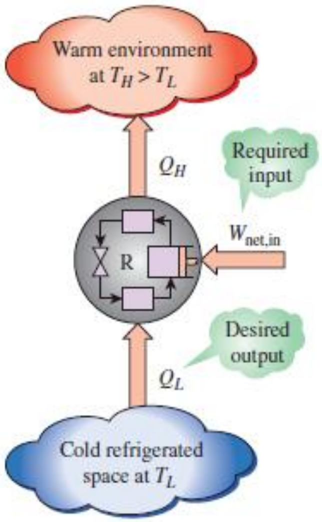
- Determining the maximum possible efficiencies of heat engines. Example; In a steam power plant, the maximum furnace temperature is 1400°C. Cooling water is available at 15°C . What is the maximum possible thermal efficiency of the thermal power plant, no matter how many refinements are included in its design?



$$\eta_{\max} = 1 - \frac{T_L}{T_H}$$

If T_L= 0K then efficiency will be 100%

- Determining the maximum possible coefficient of performance of refrigerators: Example; A refrigerator located in a room where the air temperature is 20 °C freezes 5 kg of ice per hour from water initially at 10 °C. The refrigerator rejects heat to the room. What is the absolute minimum power requirement of the refrigerator?



$$COP_{\max,R} = \frac{T_L}{T_H - T_L}$$

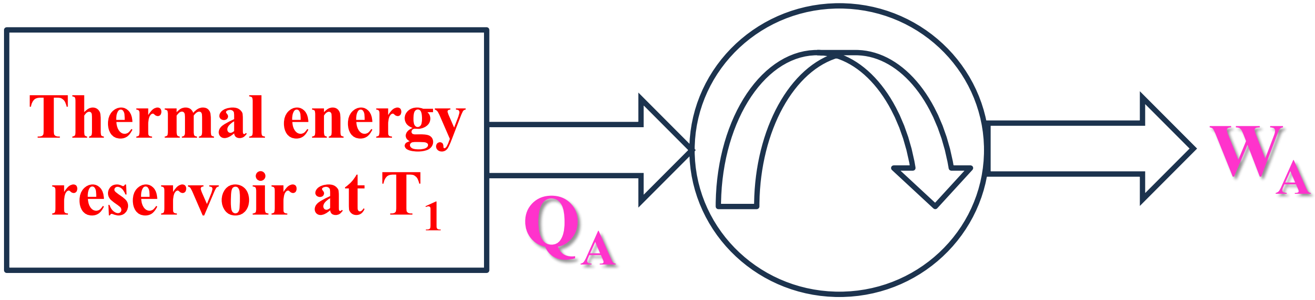
- Is there some limiting temperature below which a body can not be cooled?

Two of the major goals of the 2nd law analysis

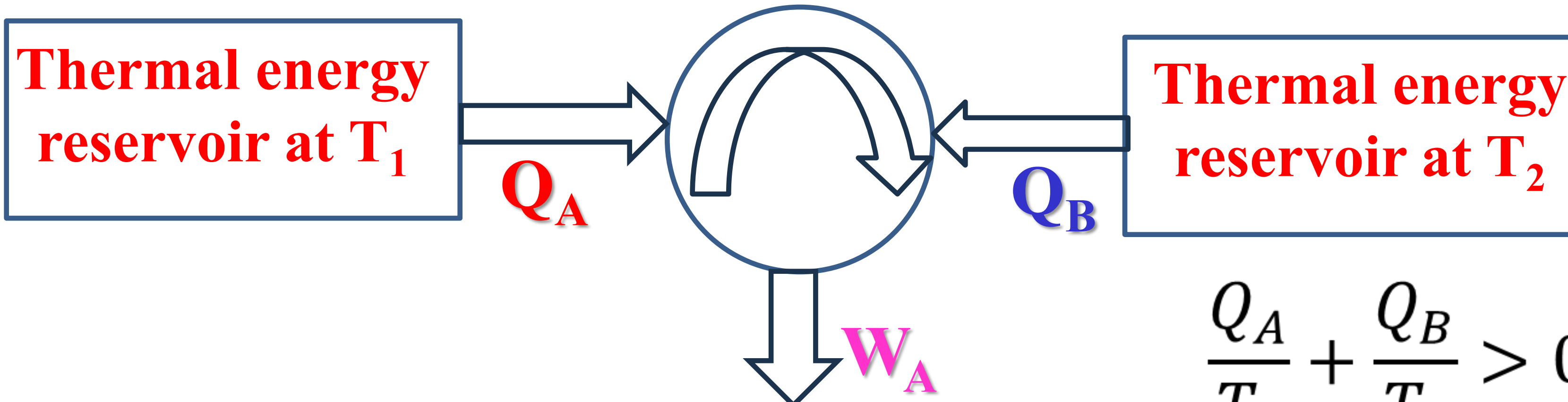
1. Determining the “ideal” or theoretically best performance of energy devices.
2. Evaluating quantitatively the factors that diminish the performance in actual practice.

Such analysis requires an understanding of the concept of reversibility and reversible process

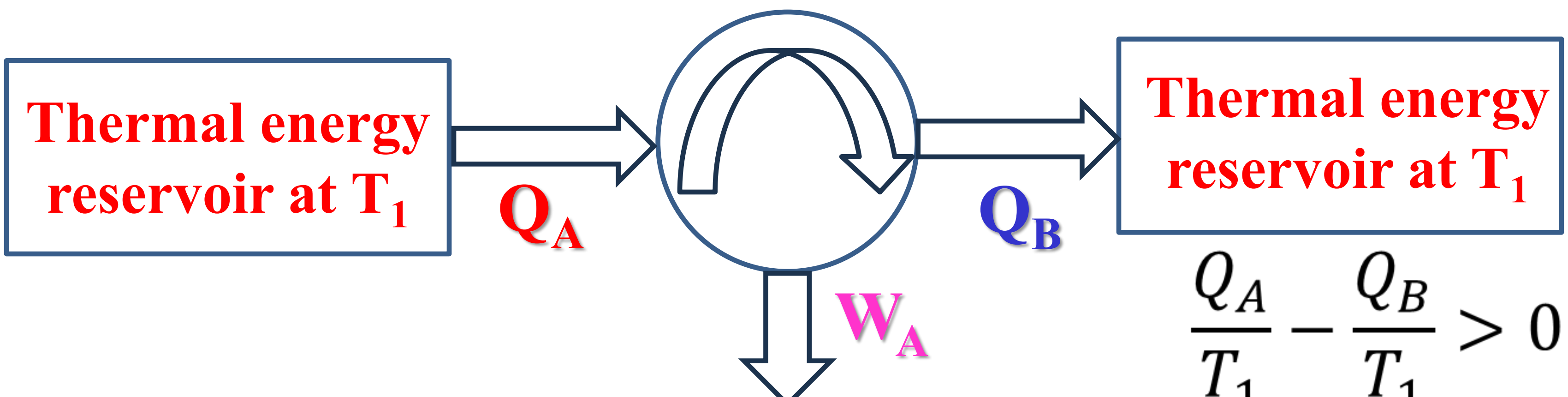
Impossible Heat Engines



$$\frac{Q_A}{T_1} > 0$$



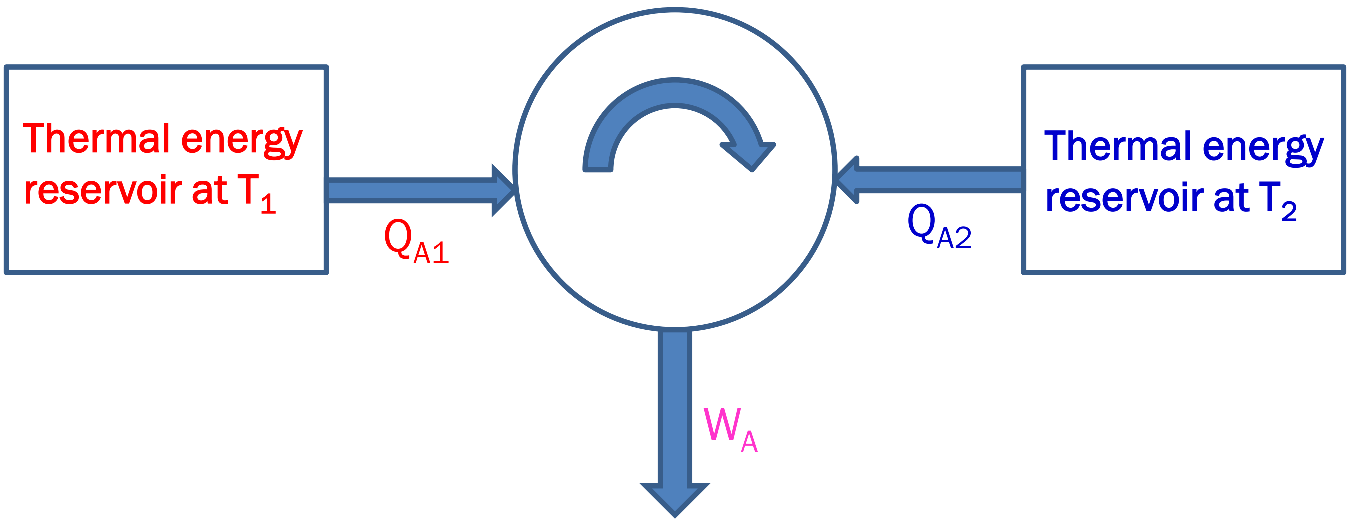
$$\frac{Q_A}{T_1} + \frac{Q_B}{T_2} > 0$$



$$\frac{Q_A}{T_1} - \frac{Q_B}{T_1} > 0$$

From constraint given by the 2nd law it has been already understood that to get continuous power output a heat engine has to interact with at least two reservoir

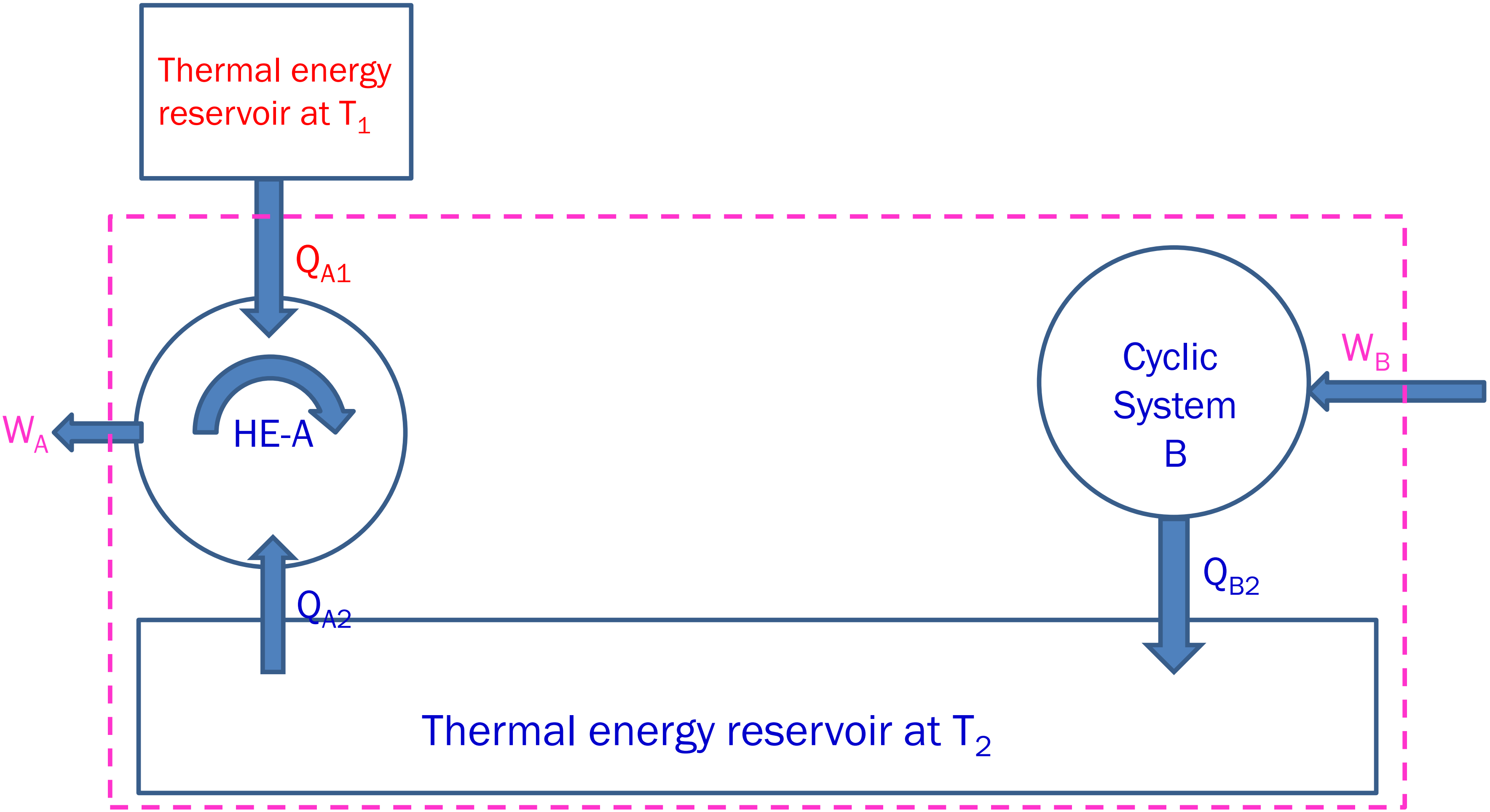
The next logical step is to determine whether a heat engine is possible which operates solely between two thermal reservoirs and whether there are sign restriction on the heat terms



The 1st law require that $$Q_{A1} + Q_{A2} - W_A = 0$$

With W_A positive there are three possibilities for the sign on the heat term

Both Q_{A1} and Q_{A2} are positive values
Both Q_{A1} and Q_{A2} are of opposite sign such that $Q_{A1} \times Q_{A2} < 0$ $\quad |Q_{A1} + Q_{A2}| > 0$



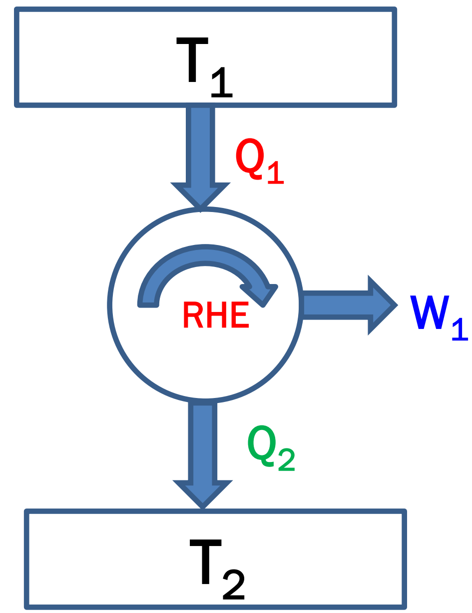
Above figure shows heat engine A operating according to option 1, where both Q_{A1} and Q_{A2} are taken into heat engine from thermal reservoir T_1 and T_2. A Cyclic device B also operates in conjunction with reservoir T_2 since only one reservoir is in contact with B, according to Kelvin-Planck statement both W_B and Q_B must be negative. Now A and B under goes integral number of cycles. So Q_{A2}=-Q_{B2}. Thus reservoir T_2 also undergoes a cycle as a result the composite system of A, B and reservoir T_2 shown by the dashed line undergoes a cycle and produce net work output while exchanging heat solely with a single reservoir at T_1 This violates the Kelvin-Planck's Statement. The only assumption made in this proof is that Q_{A1} and Q_{A2}input to the engine . Therefore this assumption can not be correct and option 1 is invalid So Q_{A1} and Q_{A2} must be of the opposite sign

Can the performance of a reversible heat engine be improved by reducing the heat sink temperature below that of the ambient?

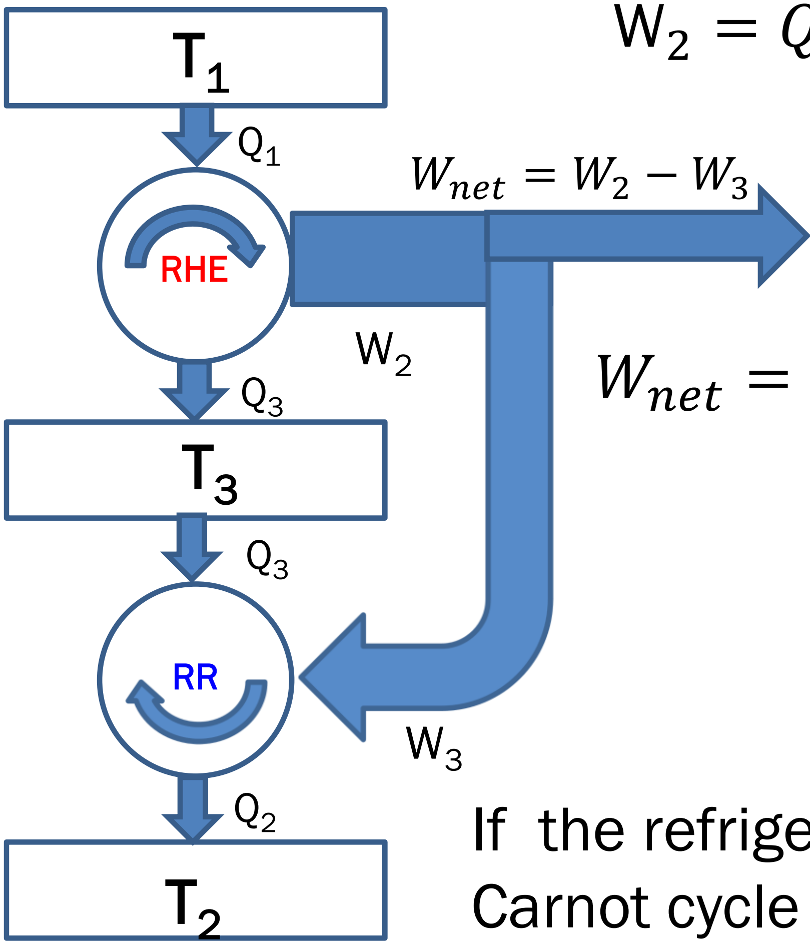
Suppose a heat engine operating between a high temperature source T_1 and the $T_2 = T_{amb}$, then can the performance be improved by reducing T_2 below that of the ambient.

$$W_1 = Q_1\left(1 - \frac{T_2}{T_1}\right)$$

$$\eta_1 = \left(1 - \frac{T_2}{T_1}\right)$$



If we want to bring the minimum temperature i.e. T_2 below that of the ambient, we need a refrigerator which maintains the sink temperature of the heat engine at $T_3 < T_2$ by extracting heat from it and rejecting to the ambient at T_2. For this external work input is required. Let's assume the refrigerator is reversible, so that the power consumption will be minimum



$$W_2 = Q_1\left(1 - \frac{T_3}{T_1}\right) \ldots\ldots 1 \qquad Q_3 = Q_1 \frac{T_3}{T_1} \ldots\ldots\ldots 2$$

$$W_3 = Q_3\left(\frac{T_2 - T_3}{T_3}\right) = Q_1 \frac{T_3}{T_1}\left(\frac{T_2 - T_3}{T_3}\right) \ldots\ldots 3$$

$$W_{net} = W_2 - W_3 = Q_1\left(1 - \frac{T_3}{T_1} + \frac{T_3}{T_1} - \frac{T_2}{T_1}\right) \ldots 4$$

$$W_{net} = Q_1\left(1 - \frac{T_2}{T_1}\right) \ldots 5$$

$$\eta_2 = \left(1 - \frac{T_2}{T_1}\right)$$

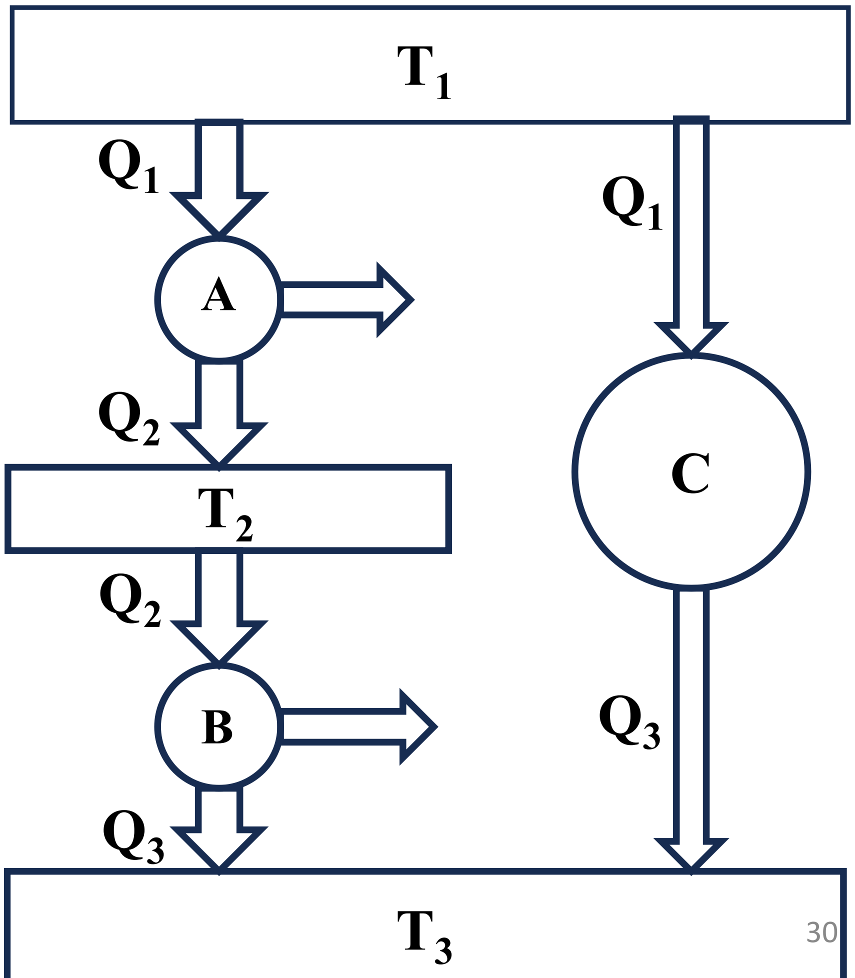
If the refrigerator does not operate on reversed Carnot cycle $\eta_2 < \eta_1$

Thermodynamic Temperature Scale

Carnot theorem show that the efficiency of a *fully reversible heat engine* operating between two thermal energy reservoir is independent of the working fluid, and is a function of temperature only. This provides a basis for temperature scale that is independent of any working substance.

According to Carnot's theorem:

$$\eta_{th} = 1 - \frac{Q_L}{Q_H} = f\left(T_L, T_H\right)$$



$$\frac{Q_1}{Q_2}=f\left(T_1,T_2\right);\frac{Q_2}{Q_3}=f\left(T_2,T_3\right);\frac{Q_1}{Q_3}=f\left(T_1,T_3\right)$$

$$\frac{Q_1}{Q_3}=\frac{Q_1}{Q_2}\times\frac{Q_2}{Q_3}\Rightarrow f\left(T_1,T_2\right)\times f\left(T_2,T_3\right)=f\left(T_1,T_3\right)$$

Since RHS is a function of T_1 & T_3

$$f\left(T_1,T_2\right)=\frac{\psi\left(T_1\right)}{\psi\left(T_2\right)};f\left(T_2,T_3\right)=\frac{\psi\left(T_2\right)}{\psi\left(T_3\right)};f\left(T_1,T_3\right)=\frac{\psi\left(T_1\right)}{\psi\left(T_3\right)}$$

For Reversible Cycle $\quad \dfrac{Q_L}{Q_H}=\dfrac{\psi\left(T_L\right)}{\psi\left(T_H\right)}$

$\psi\left(T_L\right)=T_L^n$ Is one of the several forms

The simplest of the function is with n=1 proposed by Lord Kelvin

$$\frac{Q_L}{Q_H}=\frac{T_L}{T_H}\quad or\quad \eta_{th,rev}=1-\frac{Q_L}{Q_H}=1-\frac{T_L}{T_H}$$

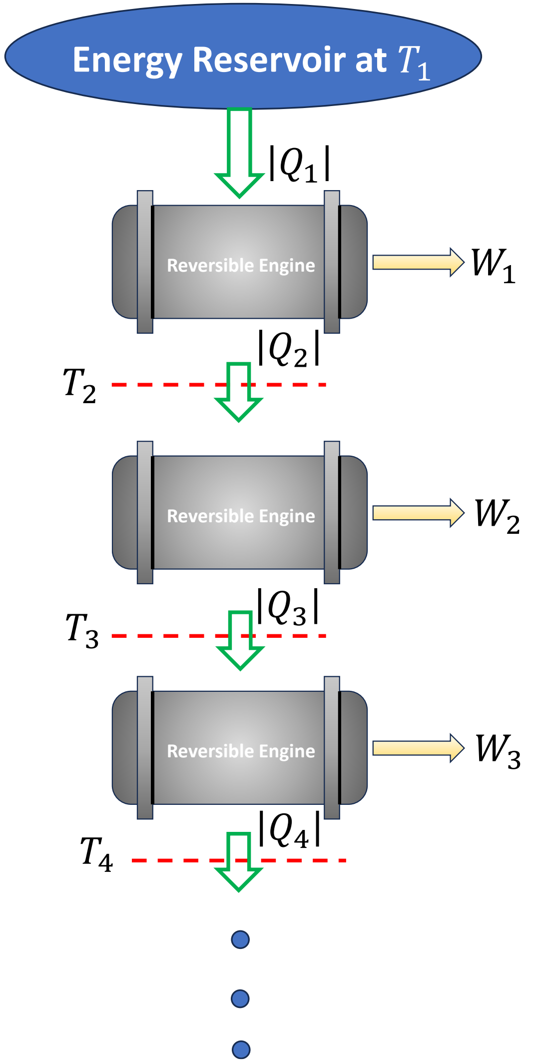
$$\frac{Q_L}{Q_H} = \frac{T_L}{T_H} \quad or \quad \eta_{th,rev} = 1 - \frac{Q_L}{Q_H} = 1 - \frac{T_L}{T_H}$$

On this scale the Kelvin scale, the ratio of the two temperatures is equal to ratio of the amount of heat transferred between the reversible heat engine and two reservoirs at these temperature

Zero on Kelvin Scale

Consider series of reversible of heat engine as shown

One engine absorbs heat $|Q_1|$ from the reservoir at T_1 it rejects heat $|Q_2|$ at temperature T_2 to another reversible engine that in tern rejects heat $|Q_3|$ at temperature T_3 to another reversible heat engine and so forth. Let the temperature be selected such that each engine does the same amount of work



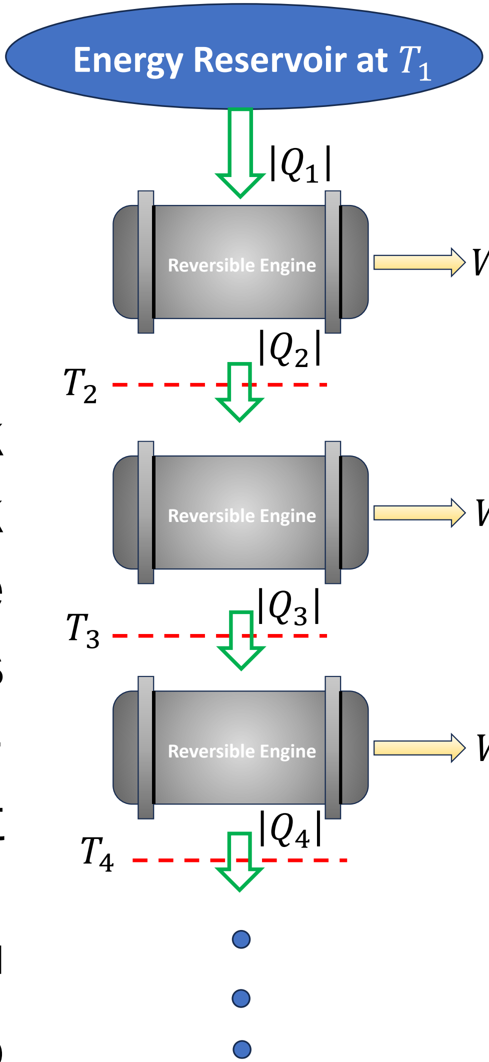
$$W_1 = W_2 = W_3 = \cdots \ldots \ldots.$$

$$|Q_1| - |Q_2| = |Q_2| - |Q_3| = |Q_3| - |Q_4| = \ldots$$

Application of defining equation of Kelvin scale shows that

$$T_1 - T_2 = T_2 - T_3 = T_3 - T_4 \ldots \ldots \ldots.$$

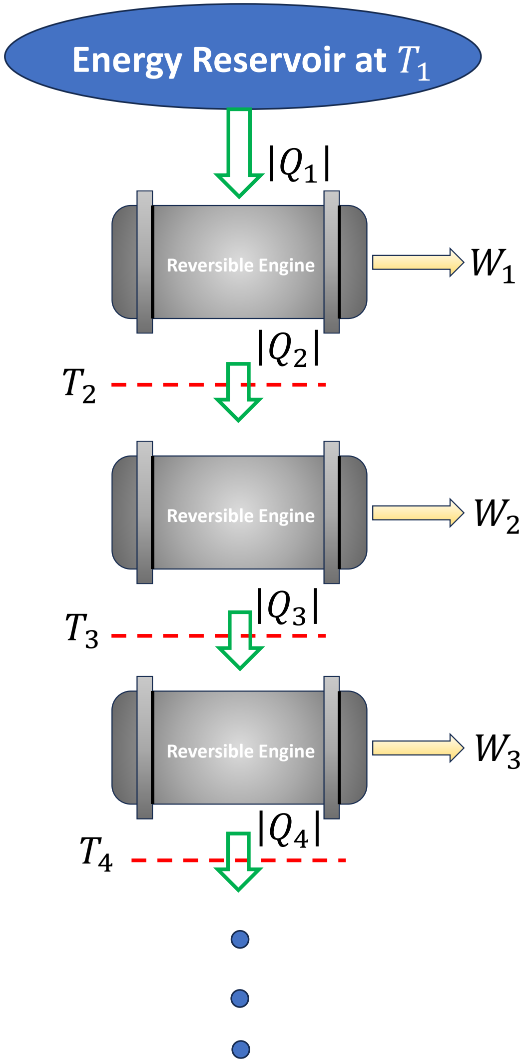
As more engines are added in series, the work output increases. By the 1st law, the total work input can not exceed $|Q_1|$. In the limiting case where there are enough engines in the series to make the total work output $W_1 + W_2 + W_3 \ldots$ equal to $|Q_1|$ the last engine will reject zero heat. The last engine in the series then violates the Kelvin-Planck statement of the 2nd law. Thus the operation of the series with zero heat rejection in the last engine can not be accomplished, although it can be approached as a limiting case.



As the number of engines in the series is increased and the heat rejection of the last engine approaches zero, the temperature at which heat is rejected also approaches zero on the Kelvin scale in accordance with the defining equation

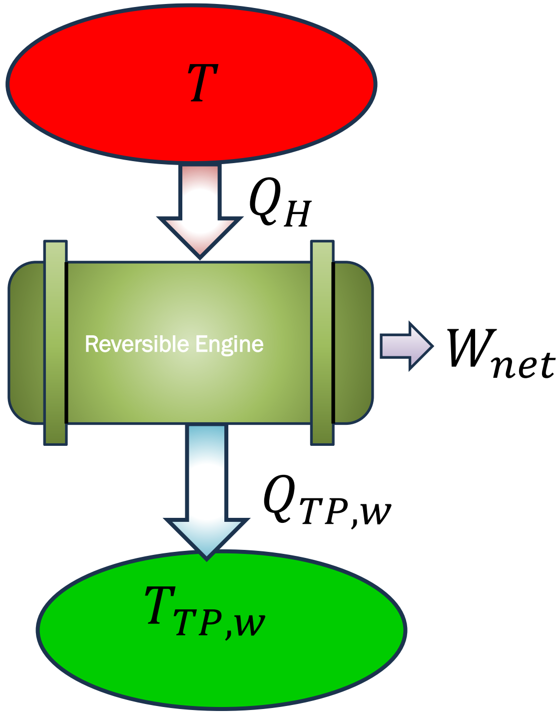
$$\frac{|Q_H|}{|Q_L|} = \frac{T_H}{T_L}$$

Absolute Zero Temperature: can be defined as follows: If a reversible heat engine operates between two reservoirs, absorbing a constant heat input from a hotter reservoir, and the temperature of the colder reservoir is successively lowered, the amount of heat rejected decreases. As the amount of heat rejected approaches zero, the temperature of the colder reservoir approaches absolute zero.



The Kelvin or Thermodynamic temperature scale is not completely defined until the unit is established. "The Kelvin unit of thermodynamic temperature, is the fraction 1/273.16 of the thermodynamic temperature of the triple point of water". Thus the numerical value in Kelvin for any other temperature is given by

$$T = T_{TP,w}\frac{|Q_H|}{|Q_{TP}|} = 273.16\frac{|Q_H|}{|Q_{TP}|}$$



Where, $|Q_H|$ and $|Q_{TP}|$ are the amount of heat transferred by a reversible engine operating between a reservoir at T and a reservoir at T_{TP}, the triple point temperature of water

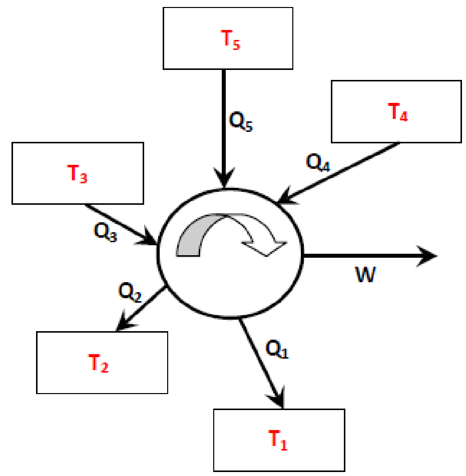
Clausius Inequality: Second law introduces the famous inequality i.e. the Clausius inequality applicable for thermodynamic cycles

Clausius inequality: $$\oint \left(\frac{\delta Q}{T}\right)_{boundary} \le 0$$

< for irreversible cycle
= for reversible cycle
> Impossible cycle

T is the absolute temperature at the part of the boundary where, δQ amount of heat transfer occur. The integration is to be performed over all points of boundary and over the entire cycle

Mathematical Definition of 2nd Law of Thermodynamics for a Cycle

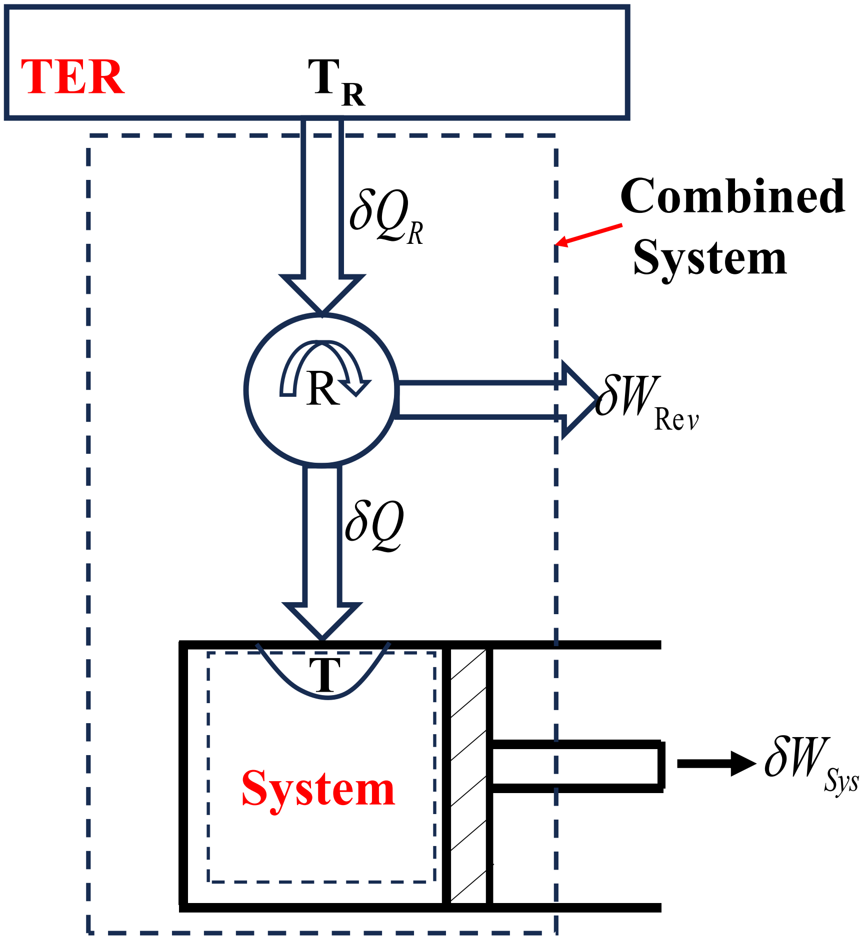


Validity of Clausius Inequality

Consider a system connected to a thermal energy reservoir (at T_R) through a reversible cycle R

The cycle receives δQ_R amount of heat from TER and supplies δQ amount of heat at temperature ***T*** to system and develops δW_{rev} amount of work

The system produces δW_{sys} amount of work as a result of this heat transfer



Applying 1st law to the Combined System (Heat engine and system)

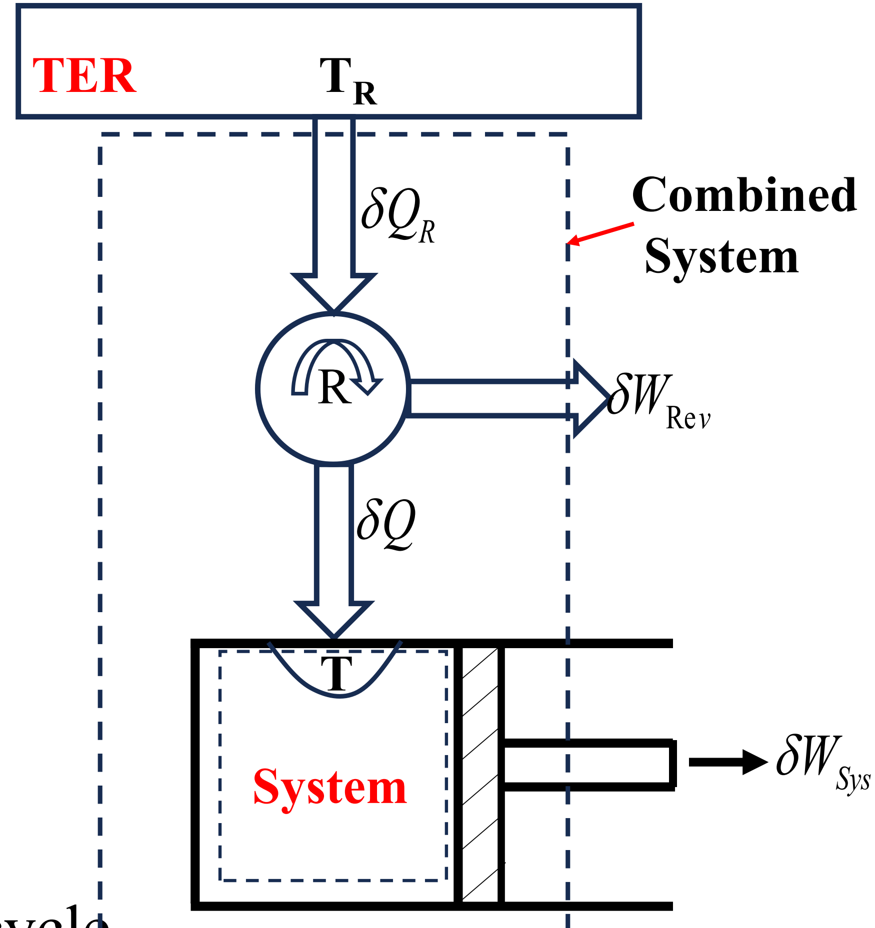
$$\delta Q_R = \delta W_C + dE_C \ldots.1$$

For reversible cycle from Carnot's theorem

$$\frac{\delta Q_R}{T_R} = \frac{\delta Q}{T} \ldots.2$$

From 1 and 2 we get

$$\delta W_C = T_R \frac{\delta Q}{T} - dE_C \ldots.3$$



Now let the system undergo a cycle while the cyclic device under goes a number of cycle

$$dE_C = 0 \qquad \delta W_C = T_R \oint \frac{\delta Q}{T} \ldots.4$$

If δW_C is positive it would violate the 2^{nd} law of thermodynamics

Kelvin-Planck Statement: $\delta W_C <= 0$ For interaction with single reservoir

$$\oint \frac{\delta Q}{T} \leq 0 \quad \left(\because T_R > 0\right)$$

The above inequality is valid for all cycles reversible/irreversible, Heat engine/Refrigerator/Heat Pumps

If no irreversible occur within the system as well as the cyclic device then the combined system will be internally reversible

The combined system can be reversed, for the reversed cycle all the quantities will have the same magnitude but opposite sign

If for the original cycle δW_C is negative then for the reversible cycle δW_C becomes positive , which violates the 2 nd law.

For the original cycle δW_C should be zero

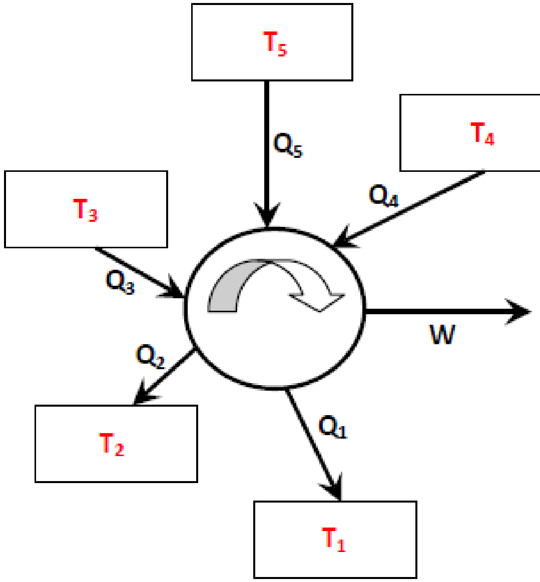
$$\Rightarrow \delta W_{C,\mathrm{int}\,rev} = T_R \oint \left(\frac{\delta Q}{T}\right)_{\mathrm{int},rev} = 0$$

$$\Rightarrow \oint \left(\frac{\delta Q}{T}\right)_{\mathrm{int},rev} = 0$$

For reversible or internally reversible cycle

$$\oint \left(\frac{\delta Q}{T}\right)_b \leq 0$$

< For irreversible
= For internally reversible



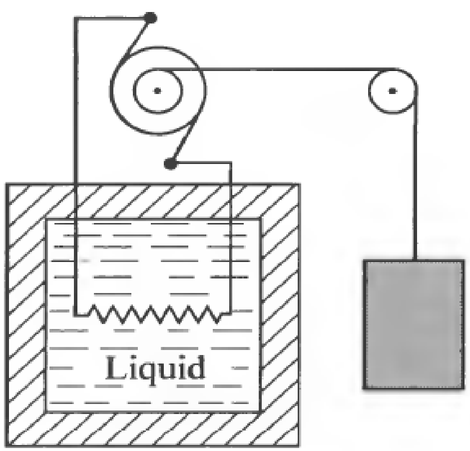
$$\sum_{i=1}^{n} \frac{Q_i}{T_i} \leq 0$$

Caratheodory Axiom-I

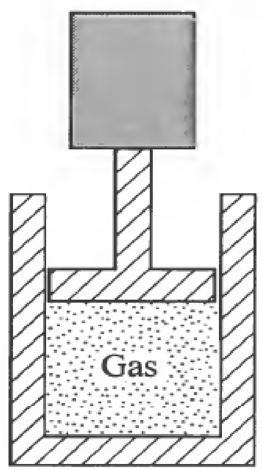
If a closed system undergoes an adiabatic change of state, the work transfer experienced by the system is the same for all adiabatic paths connecting the same initial and final states i.e. the total value of work interaction causing the change of state depend only on the end states and not on the details of the process

In this case work is used to define the change in energy E of a closed system thus

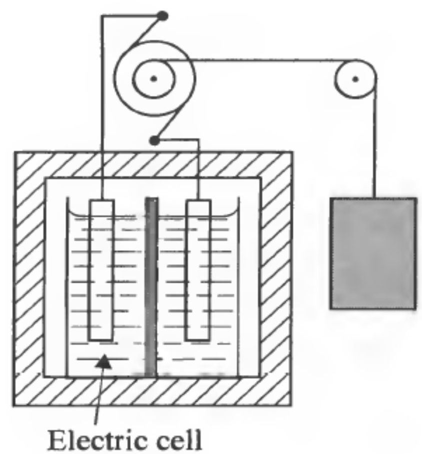
$$\delta W = -dE$$



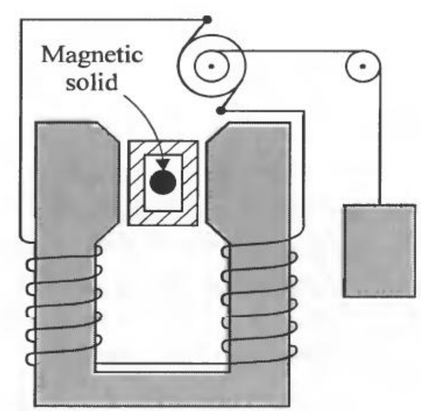
Adiabatic electrical work



Adiabatic displacement work



Adiabatic electro chemical work

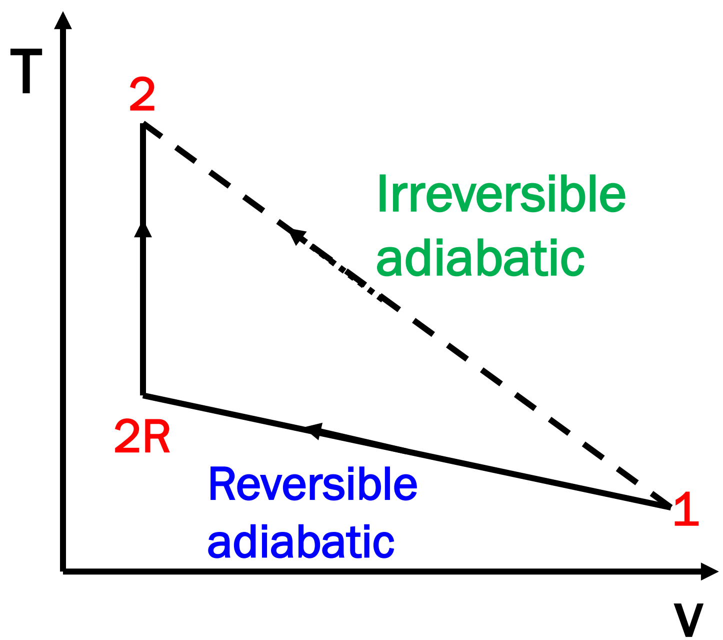


Adiabatic Magnetic work

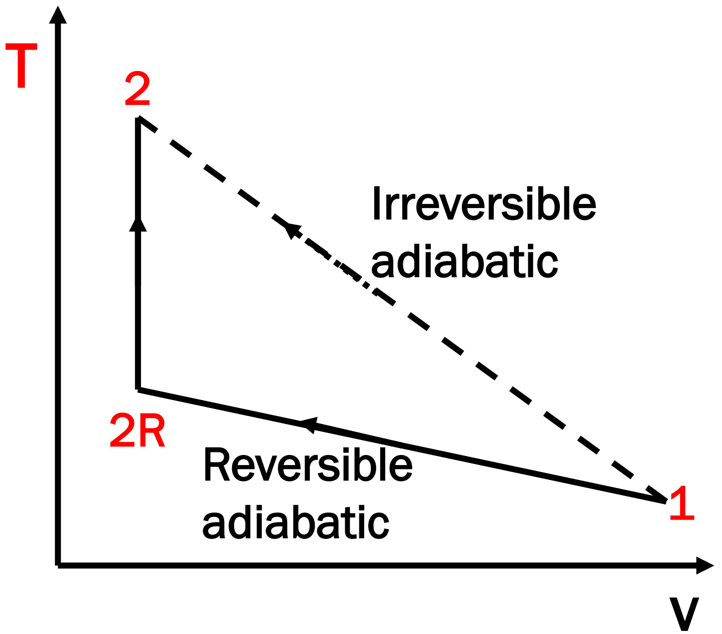
Caratheodory Axiom-II

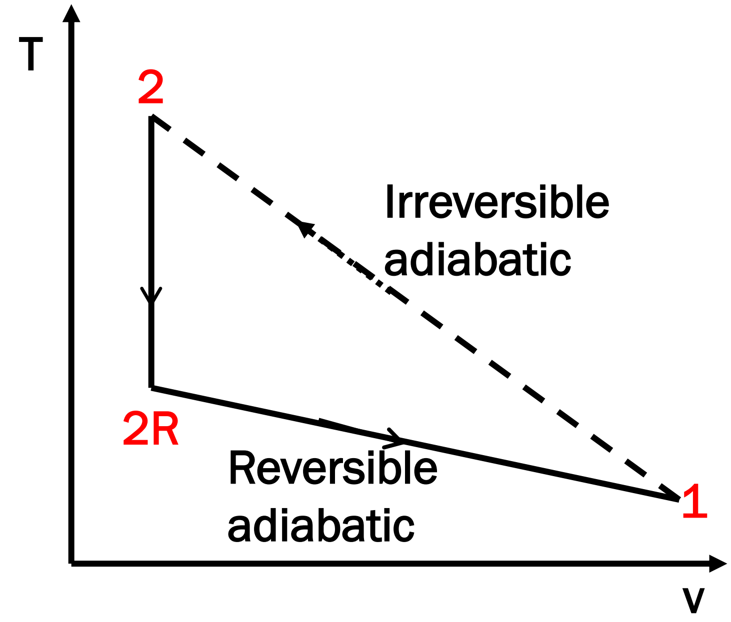
Consider the adiabatic, but irreversible compression process from (V_1 to V_2) . The same state-2 can be reached by adiabatic reversible compression 1-2R and then via heat transfer at constant volume (V_{2R}=V_2)

Is $T_{2R} > T_2$ Or $T_2 > T_{2R}$



Although V_2=V_{2R} the Caratheodory axiom-ii postulates that T_{2R} must always be lower than T_2





Proof: Assume that the axiom to be true. Since the process 1-2R-2 is reversible , one can construct the cycle 1-2-2R-1 (which contains an irreversible process 1-2). For the cycle

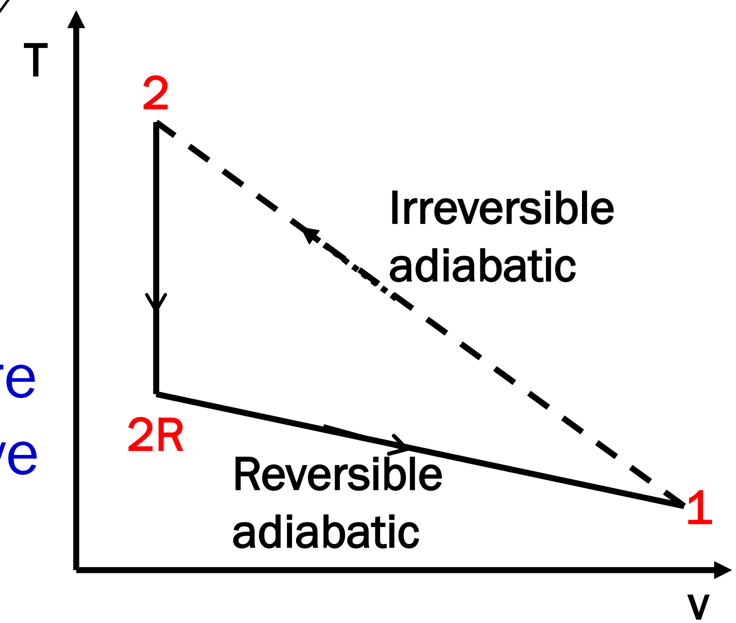
$$\oint \frac{\delta Q}{T} = \int \left(\frac{\delta Q}{T} \right)_{1-2} + \int \left(\frac{\delta Q}{T} \right)_{2-2R} + \int \left(\frac{\delta Q}{T} \right)_{2R-1} =$$

$$0 + \text{negative number} + 0 = \text{negative number (Clausius Inequality is Satisfied)}$$

The 1st and the 3rd terms are zero as these two processes are adiabatic. However, the second term is negative. How?

Process 2-2R is reversible constant volume process, where the temperature drops. For reversible process: (both for closed as well as steady flow system)

$$\delta Q = dU + pdV$$



For constant volume process: $\delta Q = C_V dT$

From the above equation if the temperature drops the heat interaction must be negative (From the system)

Now assume that the axiom is incorrect and that the state 2R lies above -2 on T-v diagram. For the cycle 1-2-2R-1

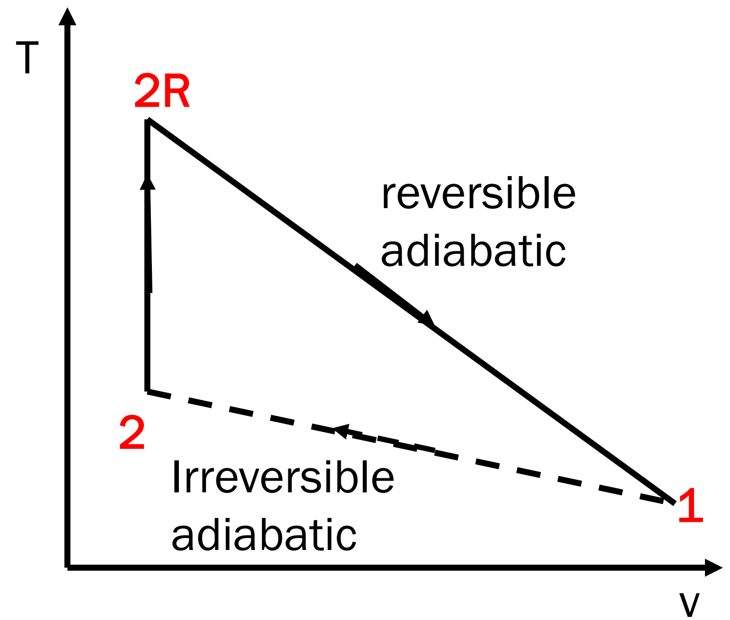
As $T_{2R} > T_2$ $\mathit{Hence}\ \delta Q_{2-2R} = C_V\, dT\ \ \mathit{is} + \mathit{ve}$

$$\oint \frac{\delta Q}{T} = \int \left(\frac{\delta Q}{T}\right)_{1-2} + \int \left(\frac{\delta Q}{T}\right)_{2-2R}$$

$$+\int \left(\frac{\delta Q}{T}\right)_{2R-1} =$$

$0 + \text{positive number} + 0 = \text{positive number}$

(Clausius Inequality is Violated)



Hence some states can not be reached through an adiabatic process, this is the essence of Caratheodory axiom -II